Flood Risk Mapping for De-mining Efforts in Iraq

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Abstract

Landmines and unexploded ordnance pose a persistent threat in post-conflict regions such as Iraq, particularly in the Diyala Governorate, which remains contaminated due to decades of war. Coupled with seasonal flooding, these hazards present a significant risk to civilian populations and demining operations. This thesis explores the integration of flood hazard mapping using Geographic Information Systems with demining efforts in Iraq, aiming to enhance the safety and effectiveness of mine clearance activities. Using the Analytic Hierarchy Process, a multi-criteria decision-making technique, and incorporating flood hazard factors such as elevation, land cover, slope, precipitation, and proximity to streams, this case study generates flood hazard maps to identify high-risk areas. These maps prioritize regions where landmine displacement due to flooding is most likely, helping to guide resource allocation for demining agencies. The findings suggest that integrating flood hazard data with existing minefield information can significantly improve the planning and execution of demining operations, reducing the risk to both civilians and deminers in flood-prone areas.

Contents

1 Introduction

Landmines have long been recognized as a grave humanitarian concern, especially in post-conflict regions where they continue to pose a significant threat to civilian populations. These explosive devices, often buried and forgotten after conflicts end, can remain active for decades, leading to accidental detonations that cause severe injuries, loss of life, and long-term socio-economic disruption. According to the *International Campaign to Ban Landmines (ICBL)*, over 60 countries worldwide are still contaminated by landmines and *Unexploded Ordnance (UXO)*, affecting the daily lives of millions of people (Ban Landmines (ICBL), 2023).

In response to the widespread impact of landmines, the international community came together to create the *Anti-Personnel Mine Ban Convention (APII)*, commonly known as the Ottawa Treaty, which was adopted in 1997 (United Nations, 1996). The treaty represents a landmark in the global effort to eliminate the use, production, and transfer of anti-personnel mines, and to assist in the clearance of existing mines. As of 2024, 164 countries have signed the treaty, committing to not only cease the use of landmines but also to engage in demining activities and provide support for mine victims. The convention has significantly reduced the number of new landmine casualties and has spurred extensive global demining operations (United Nations, 1996; Ban Landmines (ICBL), 2023).

Iraq ranks among the most heavily mine-contaminated countries globally, burdened by the remnants of decades-long conflicts, including the Iran-Iraq War, the Gulf War, and more recent hostilities (UNMAS, 2024). The extensive presence of landmines and UXO has rendered vast swathes of land unsafe for habitation or agriculture, posing ongoing dangers to local communities (JMU, 2003). In 2022 alone, these remnants injured or killed 169 individuals, 45 of whom were children, though the true number of casualties is likely higher due to under-reporting (Monitor, 2022). The persistent threat of landmines severely hampers economic development, restricts access to essential resources, and complicates the return of displaced populations (JMU, 2003). The United Nations has documented approximately 2,733 square kilometers of contaminated land, with explosive remnants affecting nearly every aspect of life in these regions (UNMAS, 2024).

To address this critical issue, the Iraqi government, in collaboration with international organizations such as the *United Nations Mine Action Service (UNMAS)* and various *Non-Governmental Organizations (NGOs)*, has undertaken significant demining efforts. These operations involve both manual and mechanical clearance of landmines and UXOs, alongside risk education programs aimed at local communities. Key initiatives include the use of advanced technology for detection and clearance, as well as the training of local demining teams. Despite the progress made, ongoing challenges such as limited funding, difficult terrain, and continued security concerns have slowed the pace of demining, leaving many areas still heavily contaminated (UNMAS, 2024).

Efficient and effective demining is crucial not only for the immediate safety of civilians but also for the long-term stability and development of Iraq. Clearing landmines enables the safe return of displaced populations, the restoration of agricultural lands, and the rebuilding of infrastructure. Moreover, it is essential for the broader process of post-conflict recovery peacebuilding, allowing communities to move beyond the legacy of war and towards sustainable development (Unruh et al., 2003).

However, the challenges of demining are further exacerbated by environmental factors, particularly flooding. Flooding presents a unique and serious complication for demining operations. Floodwaters can dislodge and transport landmines and UXOs, spreading them to previously cleared or uncharted areas and increasing the risk to both civilians and demining personnel. For instance, incidents in countries like Bosnia and Herzegovina have shown how seasonal floods can redistribute mines, making it even more difficult to track and clear these dangerous remnants (Trumble, 2021). In Iraq, where both riverine and flash flooding are common, such risks are amplified, particularly in regions already burdened by heavy landmine contamination (Shareef et al., 2021; AL-Hussein et al., 2022).

Geographic Information Systems (GISs) have become an invaluable asset in flood risk mapping, particularly when integrated with remote sensing technologies. The combination of GIS and satellite data enables the precise identification and assessment of flood-prone areas, which is crucial in regions with limited or outdated data (Wang et al., 2018). This approach is highly effective in enhancing the safety and efficiency of demining operations. For instance, GIS allows for the creation of detailed *Flood Hazard Maps (FHMs)* that can guide demining efforts by pinpointing areas most vulnerable to flooding and thus most at risk for landmine displacement, thereby reducing the dangers posed to both deminers and local populations.

This thesis represents the first step in the effort to integrate flood risk mapping into demining operations in Iraq, specifically focusing on the Diyala Governorate. The case study aims to classify high-risk flood areas to help demining agencies allocate resources more effectively and implement preventative measures to protect both deminers and local populations. The significance of this research lies in its potential to enhance demining practices in Iraq and similar contexts. By merging flood risk mapping with demining operations, this study seeks to establish a method for enhancing the efficiency and safety of landmine clearance in flood-prone areas. The findings could influence policy decisions, guide the allocation of demining resources, and contribute to the development of more comprehensive and resilient demining strategies. Ultimately, this thesis could accelerate post-conflict recovery, and support the long-term stability and development of regions affected by both landmines and flooding.

2 Literature Review

Flood hazard mapping involves identifying and delineating areas prone to flooding based on a variety of physical, environmental, and hydrological factors. Accurate flood hazard maps are critical for disaster management, as they provide a spatial representation of areas at risk, helping authorities prioritize mitigation measures and resources (Mudashiru et al., 2021). Flood hazard mapping methodologies have evolved significantly over time, with advancements in technology allowing for more sophisticated approaches. A key development is the integration of GIS and remote sensing with traditional modeling techniques (Zhou et al., 2021; Jahanbani et al., 2024). GIS enables spatial visualization of flood hazards by combining various data layers such as *Digital Elevation Models (DEMs)*, land cover, precipitation, and hydrological data into a cohesive map. Remote sensing technologies provide high-resolution data on critical variables like topography, precipitation, and elevation, which are essential for flood risk assessments (Farhadi et al., 2021; Yiran et al., 2024).

Traditionally, three main approaches are used: physically-based models, empirical methods, and physical models (Bellos, 2012; Teng et al., 2017; Mudashiru et al., 2021). Initially, physical models were widely applied but involved complex and time-consuming experiments to simulate flood behavior in real-world environments (Farhadi et al., 2021). Over time, physically-based models have largely supplanted these physical models, eliminating the need for field experiments. These models are more versatile and allow for simulations using detailed hydrological and topographical data without the need to physically recreate flood scenarios (Mudashiru et al., 2021).

Physically-based models rely on extensive datasets, including riverbed geometry, as well as hydrological and topographic parameters. These inputs are crucial for creating accurate simulations, yet their effectiveness is often constrained by the availability and quality of data. Capturing the complex dynamics of flood flow remains a challenge, particularly in areas with intricate hydrological features. Additionally, the high computational demands of physically-based models make them less accessible for regions with limited technological resources (Mudashiru et al., 2021; Shuaibu et al., 2022).

In areas where high-quality or recent data is sparse, empirical methods offer a more practical alternative (Yodying et al., 2019; Shuaibu et al., 2022). These approaches require fewer data inputs, such as DEMs and hydrological and topographic data, and are more adaptable in data-limited contexts. Empirical methods can be broadly categorized into several types, including *Multi-Criteria Decision-Making (MCDM)* techniques, statistical models, and machine learning-based approaches (Mudashiru et al., 2021). While empirical methods may not provide the same level of precision as physically-based models, they are often sufficient for preliminary risk assessment, especially in under-resourced regions.

Among the empirical methods, MCDM techniques like the *Analytical Hierarchy Process (AHP)* have proven particularly useful for flood risk assessment (Chen et al., 2014; Yiran et al., 2024). Developed by Thomas Saaty in the 1980s, AHP is a structured decision-making technique that organizes and evaluates the relative importance of various criteria affecting flood risk (Saaty, 1980). AHP is especially valuable for its ability to incorporate both qualitative and quantitative data, making it ideal for situations where diverse factors, such as topography, precipitation, soil type, and land cover, need to be assessed (Chen et al., 2014; Yodying et al., 2019; Ekeanyanwu et al., 2022; Shuaibu et al., 2022).

The AHP process begins by establishing a hierarchical structure of flood risk criteria, with the overall goal of identifying flood-prone areas at the top and specific contributing factors like slope, soil type, and precipitation arranged below it. Pairwise comparisons are then used to evaluate the relative importance of each factor, producing a matrix from which numerical weights are derived. These weights are used to generate flood risk maps that reflect the significance of each factor (Saaty, 1980; Chen et al., 2014).

In practice, AHP, when combined with GIS, has been applied successfully in various contexts. For instance, in the Hadejia River Basin in Nigeria, Shuaibu et al. used AHP and GIS to assign weights to both hydro-geomorphic factors, such as rainfall, elevation, slope, soil type, and drainage density, as well as socio-economic indicators like population density, literacy rate, and land use. This approach resulted in a flood hazard map that offered valuable insights into regional flood risks across the region (Shuaibu et al., 2022). Similarly, in Thailand's Bang Rakam Model 60 project area, Yodying et al. applied AHP to assess flood hazards by evaluating eight key factors, including flow accumulation, soil water infiltration and elevation. The study revealed that areas close to drainage networks were particularly vulnerable to flooding (Yodying et al., 2019). A comparable approach was used in Austin, Texas, where Ekeanyanwu et al. employed AHP and GIS to generate flood hazard maps using factors like precipitation, slope, elevation, flow accumulation, land cover, and proximity to water bodies. This methodology provided a comprehensive flood risk assessment for the city, helping local authorities better understand flood-prone areas and prioritize mitigation efforts (Ekeanyanwu et al., 2022).

Despite its utility, AHP has limitations. The method relies heavily on expert judgment to assign weights to criteria, introducing the potential for subjective bias. Inconsistent pairwise comparisons, especially when a large number of criteria are involved, can also affect the accuracy of the results. Additionally, constructing the comparison matrix can become cumbersome and time-consuming, particularly in studies involving numerous criteria (Ishizaka, 2019).

3 Data Collection and Preprocessing

The collection and preprocessing of data are essential for ensuring accurate and reliable flood hazard modeling in the Diyala Governorate. This region, characterized by significant hydrological variability (Abdelmohsen et al., 2022), presents unique challenges due to its topography, land cover, and climatic conditions, all of which contribute to the complexity of flood risk assessment. Effective modeling in such a context is crucial not only for understanding flood hazards but also for the planning of demining operations, where accurate environmental data is vital for operational safety and efficiency.

Data on elevation, precipitation, land cover, and hydrology must be gathered, processed, and standardized to maintain the integrity of the modeling process. Any errors or inconsistencies at this stage can lead to significant inaccuracies in the final FHMs, ultimately compromising both flood risk management and the safe planning of demining efforts. This section outlines the systematic procedures employed to collect and prepare the necessary datasets for this case study, ensuring a robust foundation for the subsequent analysis.

3.1 Study Area and Shapefile

This thesis focuses on the Diyala Governorate, a semiarid region in Iraq that faces increasingly severe weather patterns due to the impacts of climate change (S. Al-Khafaji, 2019). Prolonged droughts followed by intense rainfall, and flash flooding contribute to the area's hydrological variability and present significant challenges for managing water resources and flood risk (Abdelmohsen et al., 2022). The water resources of Iraq are largely defined by its two major river systems, the Tigris and Euphrates, along with their tributaries, which play a crucial role in supporting agriculture, communities, and ecosystems (Nadhir, 2021). Diyala, being part of the Tigris River basin, is particularly influenced by these water networks. As a result, understanding the complex interaction between climate-driven weather extremes and local water resources is essential for effective flood hazard modeling and the planning of demining operations.

The Diyala Governorate, depicted in Figure 1, covers approximately 17.685 square kilometers and is traversed by the 445-kilometer-long Diyala River. A critical water source for the region, the river originates in the Zagros Mountains of Iran and flows into the Tigris River south of Baghdad City (S. Al-Khafaji, 2019). The Diyala River is fed by several tributaries, including the Sirwand, Wand, and Tangro rivers. The river's flow is regulated by the Darbandikhan Dam, located upstream in the northeastern Sulaymaniyah Governorate, and constructed between 1956 and 1961 to provide irrigation, flood control, and hydroelectric power generation (Younus et al., 2023). Further downstream, the Hamrin Dam, completed in 1981 as part of the Khalus Irrigation Project, also plays a key role in water management, supporting hydropower generation and irrigation (S. Al-Khafaji, 2019). Figure 2 illustrates the locations of these two critical dams: Point 1 marks the position of the Darbandikhan Dam, while Point 2 indicates the Hamrin Dam.

This intricate network of rivers and dams significantly influences the hydrology of the Diyala region, which experiences distinct seasonal variations in water availability. The flood season spans from October to May, with two distinct phases: a variable flood period from October to February, characterized by fluctuating rainfall intensity, followed by a more consistent flood period from March to May (Nadhir, 2021). The dry season, lasting from June to September, brings a stark contrast with minimal rainfall and reduced river flow. Given these seasonal hydrological patterns, the FHMs were generated for the months of October through May to account for the higher flood risks during this time frame (Al-Nassar et al., 2021).

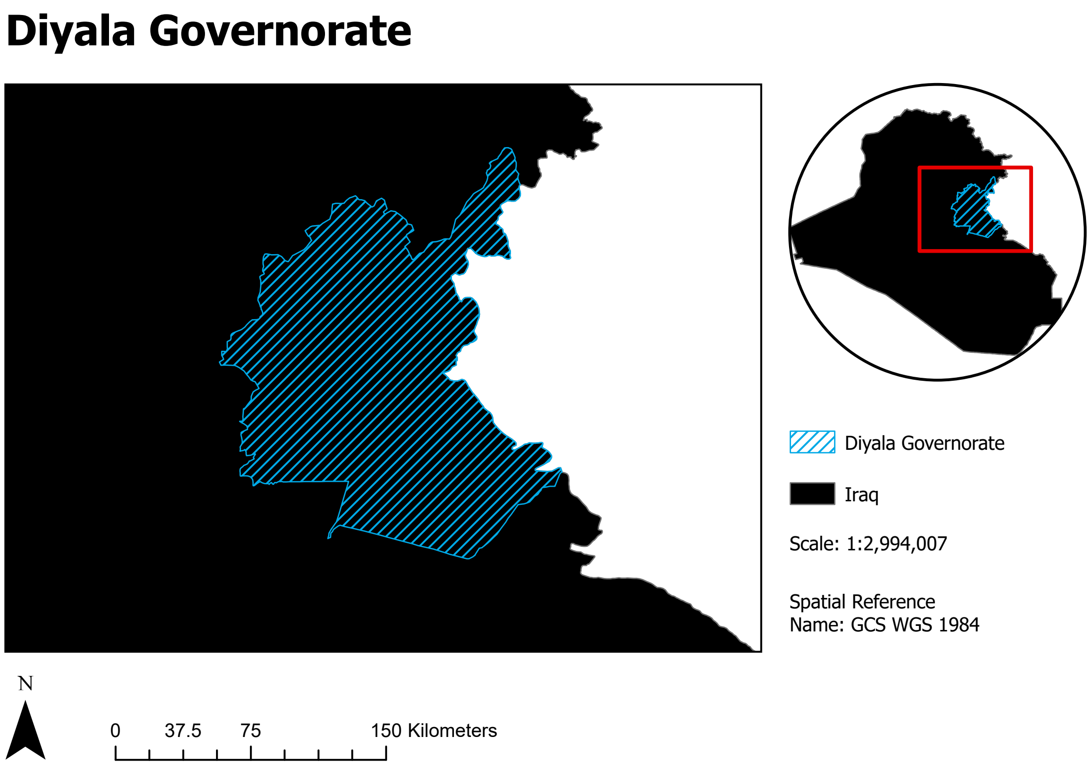

Figure 1: Map of Iraq highlighting the Diyala Governorate.

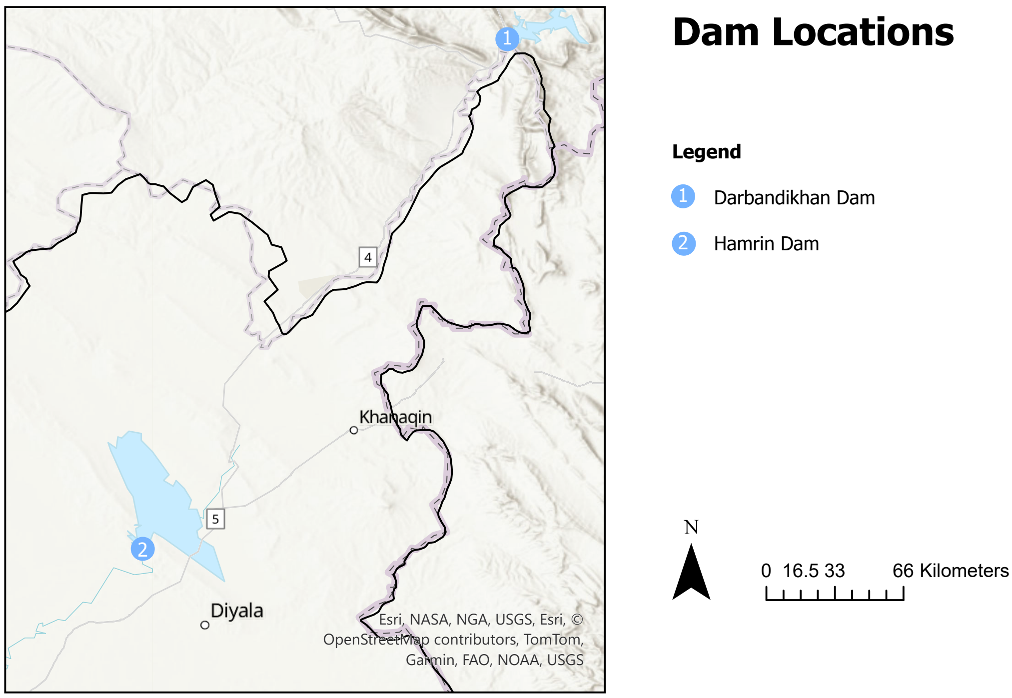

Figure 2: Map of Diyala Governorate showing the location of two major dams: Point 1 marks the Darbandikhan Dam, and Point 2 marks the Hamrin Dam.

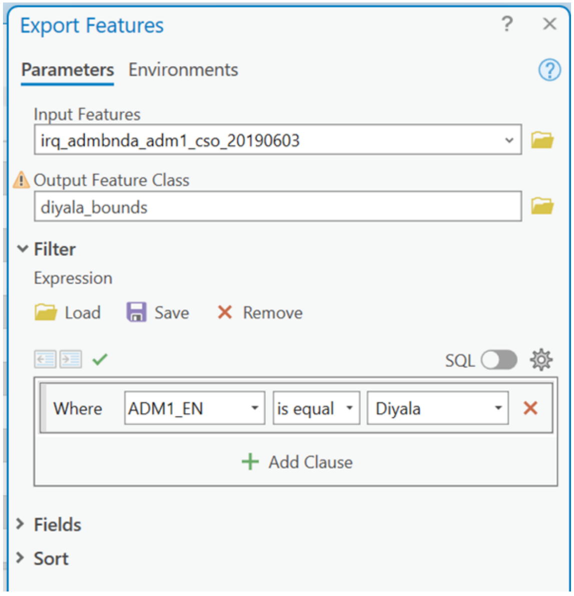

Figure 3: Clipping the Diyala Governorate from the Iraq shapefile using ArcGIS Pro. The 'adm1' shapefile was used, with a filter applied to isolate the boundary of the Diyala region.

Table 1: Administrative Levels of Iraq Shapefile. The structure of the downloaded shapefile data includes various administrative levels, from the country to sub-district, each represented by a corresponding number of geographic features (OCHA Iraq, 2019).

File Signature	**adm0**	**adm1**	**adm2**	**adm3**
Administrative Type	Country	Governorate	District	Sub-district
Features	1	19	102	294

For spatial analysis in this case study, a shapefile representing the Diyala Governorate was obtained from OCHA Iraq and processed using ArcGIS Pro. To ensure that the analysis concentrates specifically on the Diyala area, the shapefile boundaries were clipped, as illustrated in Figure 3. This clipped shapefile serves as the geographic basis for all subsequent analysis layers, thereby confining flood hazard modeling to the Diyala Governorate.

This shapefile corresponds to the "adm1" administrative level in Iraq, allowing for a focused examination of the region's flood risk. To contextualize this data, Table 1 outlines the structure of the shapefile data, which includes various administrative levels: country (adm0), governorate (adm1), district (adm2), and sub-district (adm3). The "adm1" shapefile contains 19 features, each representing one of Iraq's 19 governorates, including Diyala. By focusing exclusively on the Diyala region, the analysis streamlines subsequent layers — such as elevation, precipitation, and land cover — ensuring they are geographically aligned with the governorate boundaries. This alignment is essential for enhancing the accuracy and reliability of the flood risk assessments conducted in this case study.

3.2 Monthly Precipitation

The monthly precipitation data for this case study was sourced from WorldClim version 2.1, as provided by Fick et al. This dataset, released in January 2020 and covering the pe-

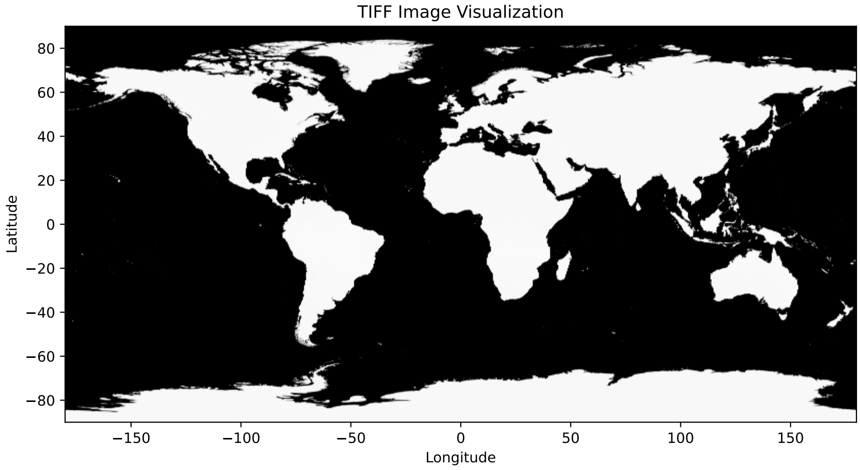

Figure 4: Precipitation Data Visualization for January, showing the global scale of the dataset.

riod from 1970 to 2000, offers high-resolution global climate data, including key variables like precipitation and temperature, which are crucial for environmental and climate-related research (Fick et al., 2017). Precipitation is integral to flood hazard mapping, as it influences surface runoff, soil saturation, and river discharge — factors that directly affect flood risks (Tabari, 2020). Accurate precipitation data helps capture rainfall patterns, essential for predicting flood events and understanding their potential impacts.

The dataset used in this thesis focuses specifically on precipitation, with a spatial resolution of 30 arc-seconds, approximately equivalent to one square kilometer. This high-resolution data is key to producing reliable FHMs because it captures the spatial variability of precipitation across the Diyala Governorate. Such granularity is particularly important in regions with diverse topography or microclimates, where precipitation can vary significantly across short distances (Pham et al., 2024). In flood hazard modeling, this level of detail allows for more precise predictions about where and when floods might occur and their potential severity. Using lower-resolution data could obscure critical local variations in precipitation, leading to less reliable FHMs and underestimating flood risks in certain areas. Therefore, high-resolution precipitation data is essential for accurate risk assessment.

As shown in Figure 4, the global precipitation GeoTIFF file from WorldClim encompasses the entire world. To focus on the Diyala Governorate, the data must be clipped to the region's boundaries before analysis can begin. This process isolates the region of interest and ensures that the analysis remains relevant to the study area. The shapefile used for clipping the precipitation data corresponds to the one described in Section 3.1.

In addition to historical precipitation data, this case study uses projections based on *General Circulation Models (GCMs)* simulating the Earth's climate system by accounting for atmospheric circulation, ocean currents, and surface-atmosphere interactions. These models are fundamental tools for predicting future climate scenarios, as they assess how factors like greenhouse gas emissions may influence future climate patterns (Stute et al., 2001). To assess future flood risks in the Diyala Governorate, this thesis employs GCM simulations incorporating the new *Shared Socioeconomic Pathway (SSP)* from the *Coupled Model Intercomparison Project 6 (CMIP6)*. SSPs describe various ways global society, demographics, and economics might evolve in the 21st century, factoring in elements like

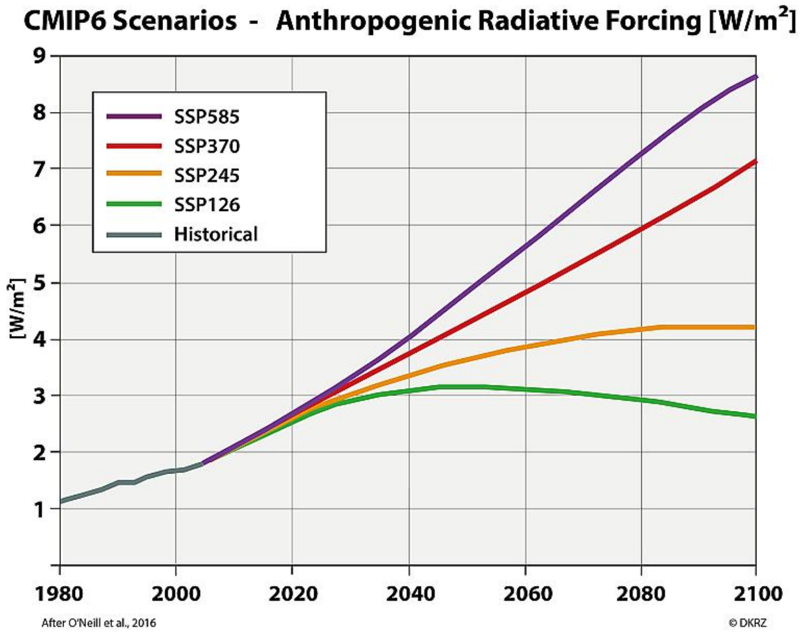

Figure 5: Projected trends in radiative forcing under different SSP scenarios. SSP585 represents the highest radiative forcing by 2100, while SSP126 envisions the most optimistic outcome with substantial climate protection measures. The color-coded lines in the figure correspond to the different SSP scenarios: SSP585 (purple), SSP370 (red), SSP245 (yellow), and SSP126 (green). Reprinted from (Böttinger et al., 2021).

population growth, economic development, technological advancements, and policy decisions related to climate change mitigation and adaptation (Böttinger et al., 2021). The scenarios span a range of potential outcomes, from high to low radiative forcing levels — a measure of factors such as greenhouse gas emissions affecting Earth's energy balance. Figure 5 illustrates the projected trends in radiative forcing for different SSPs, highlighting their implications for global CO2 concentrations and future precipitation patterns.

- **SSP585**: The most extreme scenario, projecting an additional radiative forcing of 8.5 W/m^2 by 2100. It updates the CMIP5 *Representative Concentration Pathway (RCP)* 8.5 scenario and assumes minimal climate mitigation. Shown in purple in Figure 5.
- **SSP370**: This upper-middle scenario predicts 7 W/m^2 by 2100, bridging the gap between RCP6.0 and RCP8.5. Shown in red in Figure 5.
- **SSP245**: An update to RCP4.5, this scenario assumes moderate climate protection, with radiative forcing reaching 4.5 W/m^2 by 2100. Shown in yellow in Figure 5.
- **SSP126**: The most optimistic scenario, aiming for substantial climate protection to limit radiative forcing to 2.6 W/m^2 by 2100, in line with the 2°C global temperature target. Shown in green in Figure 5.

Accurate predictions of precipitation patterns are essential for creating FHMs that address both current and future flood risks in the Diyala Governorate (Böttinger et al., 2021). For this case study, the HadGEM3-GC31-LL model was selected as the GCM, based on recommendations from the *United Nations Development Programme (UNDP)*. By incorporating precipitation data from the SSP126, SSP245, and SSP585 scenarios, the analysis captures a wide range of possible future conditions from 2021 to 2040. This comprehensive approach enhances the accuracy of the FHMs, making them more reliable for planning demining operations under various potential future climates.

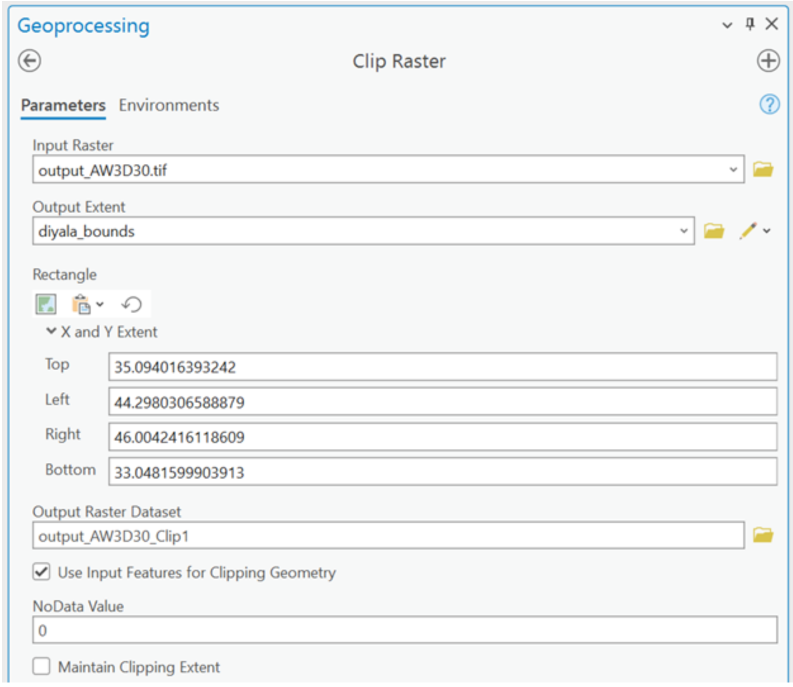

Figure 6: Interface of the clipping tool used to extract the DEM for the Diyala Governorate region.

3.3 Digital Elevation Model

The DEM utilized for this case study is the *Advanced Land Observing Satellite (ALOS)* World 3D 30-meter DEM (Japan Aerospace Exploration Agency, 2021). Although higher-resolution options exist, such as the 12-meter TanDEM-X DEM and 1-meter LiDAR datasets, the 30-meter resolution offers a great open-source alternative with good balance between detail and computational efficiency for topographic and flood hazard modeling within the Diyala Governorate (Tadono et al., 2016). In flood hazard mapping, DEMs are critical for accurately representing terrain features that influence water flow, such as slope, elevation changes, and drainage patterns. These features directly affect surface runoff, water accumulation, and potential flood zones (Oriola et al., 2016).

The ALOS World 3D DEM is derived from data captured by the ALOS using the *Panchromatic Remote-sensing Instrument for Stereo Mapping (PRISM)*. PRISM is a high-performance optical sensor capable of capturing stereo images from forward, nadir, and backward views simultaneously, which are then processed to create three-dimensional topographic models. This multi-angle imaging allows for highly accurate elevation data, making it valuable for terrain analysis in flood hazard mapping (Tadono et al., 2016).

Once the DEM data was obtained, it was integrated into a new map project using ArcGIS Pro. To ensure the data was relevant to the study area, the raster file was clipped to the boundaries of the Diyala Governorate. This clipping tool, as shown in Figure 6, takes in the unclipped DEM and the shapefile described in Section 3.1, tailoring the dataset to the specific geographic region of interest and enhancing the relevance and accuracy of the subsequent analysis.

3.4 Land Cover

The land cover dataset utilized in this case study was derived from ESA Sentinel-2 imagery, which provides a spatial resolution of 10 meters, allowing for detailed environmental

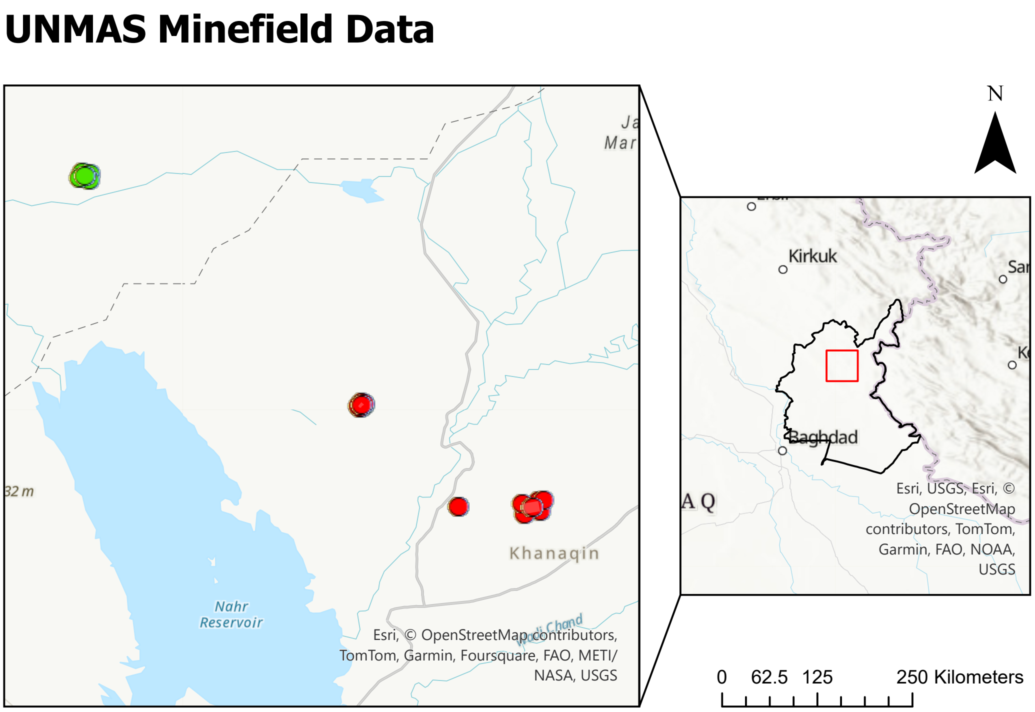

Figure 7: Map of the Diyala Governorate displaying UNMAS Hazard Data from 2019 to 2022. Red points indicate locations of mines that have not yet been cleared, while green points represent sites where mines have been successfully cleared.

analysis. This dataset was processed using Impact Observatory's advanced deep learning-based land classification model, specifically designed for accurate land cover mapping. The model was trained on a vast dataset of human-labeled images, provided by the National Geographic Society, ensuring high precision and reliability in distinguishing land cover types (Karra, 2021).

To obtain the dataset, the "Map Viewer" interface described in the dataset's documentation was accessed, and the "Download GeoTIFF" option was selected. The GeoTIFF tile was specifically chosen for the study area, focusing on the land cover layer corresponding to the year 2023. The resulting land cover classification includes the categories Water, Trees, Flooded Vegetation, Crops, Built Area, Bare Ground, Snow/Ice, Clouds, and Rangeland, as detailed in Table 2.

The land cover data was clipped to the region of the Diyala Governorate, revealing that the "Snow/Ice" and "Clouds" classes were absent from the dataset within this area. This absence is attributed to the geographic characteristics of the region. The clipping process was conducted using the same method as demonstrated in Figure 6, ensuring that the land cover raster file precisely aligns with the boundaries of the Diyala Governorate.

3.5 Minefield Data

The minefield data used for this analysis was sourced from the UNMAS, covering the years 2019 to 2022. UNMAS maintains a comprehensive dataset on mine-affected areas and clearance operations. The map data is provided in KMZ format, a compressed file type commonly used in geospatial applications like Google Earth and GIS software. KMZ files store *Keyhole Markup Language (KML)* data, which includes geographical features

Table 2: Landcover Class Definitions. Reproduced from (Karra, 2021)

Value	Name	Description
1	Water	Areas with consistent year-round water presence, excluding regions with occasional or temporary water. Typically devoid of sparse vegetation, rock outcrops, or constructed features such as docks. Examples include rivers, ponds, lakes, oceans, and inundated salt flats.
2	Trees	Significant clustering of tall, dense vegetation (approximately 15 feet or higher), usually with a closed or thick canopy. Examples include forests, dense clusters of tall vegetation in savannas, plantations, swamps, and mangroves where the canopy is too thick to detect underlying water.
4	Flooded Vegetation	Vegetation areas frequently intermixed with water throughout most of the year; includes seasonally flooded regions with a mix of grass, shrubs, trees, and bare ground. Examples are flooded mangroves, emergent vegetation, rice paddies, and heavily irrigated or inundated agricultural fields.
5	Crops	Human-planted cereals, grasses, and crops below tree height; examples include corn, wheat, soybeans, and fallow fields.
7	Built Area	Human-made structures; extensive road and rail networks; large, uniform impervious surfaces including parking structures, office buildings, and residential housing; examples: houses, dense villages/towns/cities, paved roads, and asphalt.
8	Bare Ground	Areas with minimal to no vegetation year-round, including rock and soil; extensive sand and deserts with sparse vegetation; examples are exposed rock or soil, desert dunes, dry salt flats, and abandoned mines.
9	Snow/Ice	Large, uniform regions of permanent snow or ice, typically found in mountainous areas or polar regions; examples include glaciers, permanent snowpacks, and snow fields.
10	Clouds	Areas lacking land cover information due to persistent cloud cover.
11	Rangeland	Open spaces covered predominantly by grasses with minimal tall vegetation; natural meadows, fields with sparse or no trees, open savanna with few trees, parks, golf courses, and pastures. Land with small plant clusters or individual plants on exposed soil or rock; scrub-filled clearings in dense forests that are not taller than trees; examples include moderate to sparse cover of bushes, shrubs, and tufts of grass, savannas with very sparse vegetation.

Table 3: Minefield Data

OBJECTID_1	130	**hazard_sta**	Active
hazard_pro	Diyala	**hazard_dat**	9/4/2019
hazard_dis	Khanaqin	**hazard_lat**	34.254411
hazard_cit	Wadi Awsaj	**hazard_lon**	45.209587

such as points, lines, and polygons, along with associated metadata, enabling users to visualize and analyze minefield locations and characteristics. By importing these KMZ files into ArcGIS Pro, users can perform detailed analysis, query specific locations, and filter data based on attributes such as mine status or geographic coordinates.

As shown in Figure 7, the map contains two key elements: Hazard Areas and Hazard Points. Hazard Areas are broader zones identified as mine-affected, while Hazard Points represent specific locations of individual mines. The red points denote uncleared mines, and green points represent sites where clearance has been completed. This visual distinction is crucial for operational planning and prioritizing demining efforts.

In ArcGIS Pro, users can select hazardous points to access detailed information. For example, Table 3 presents data for a specific mine (OBJECTID_1 = 130), including its status, geographic location, and the date it was recorded. In Table 3, object ID 130 refers to an active mine located in the Khanaqin District of Diyala. The "Active" status (hazard_sta) indicates that the mine has not been cleared, while the date (hazard_dat) shows when the hazard was recorded. Precise coordinates (hazard_lat and hazard_lon) allow field teams to locate the mine for safe and effective clearance. However, as noted in Section 5.4, multiple mines within the same Hazard Area are assigned identical coordinates, despite representing distinct points on the map. This could be a result of generalizing the mines under a common Hazard Area, or it may reflect incomplete or approximate data on individual mine locations. Addressing these data gaps is crucial for improving the precision of demining operations and ensuring that no mine is overlooked.

In the final step of this case study, the Hazard Areas and Points are overlaid with the generated FHMs. This integration allows users to identify mines located in flood-prone areas and prioritize their clearance, as flooding could potentially displace mines or increase the risk of accidental detonation. Integrating minefield and flood hazard data provides a strategic advantage by identifying regions where demining efforts are most urgent due to compounded risks, such as mine displacement during flooding. This approach ensures that clearance operations prioritize high-risk zones, safeguarding both the local population and demining personnel.

4 Methodology

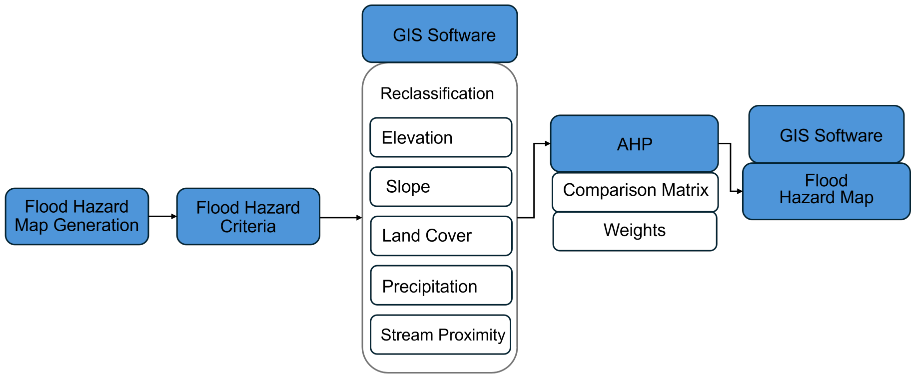

Figure 8: Conceptual framework illustrating the methodology for generating Flood Hazard Maps in Diyala Governorate.

The methodology begins by identifying key flood hazard criteria, selected for their relevance to flood risk and the desired level of accuracy. For this case study, the chosen criteria — elevation, slope, land cover, precipitation, and proximity to streams — were deemed critical for assessing flood vulnerability in the Diyala Governorate, as shown in Figure 8. Similar criteria haven been used in previous studies conducted by Shuaibu et al., Yodying et al. and Ekeanyanwu et al. Once these criteria were established, relevant data were imported into ArcGIS Pro. The data layers were reclassified and combined using weights derived from the AHP, ensuring a data-driven and systematic approach to flood hazard mapping.

4.1 DEM Modifications

The main goal of modifying the DEM in this case study is to generate the stream proximity layer, a critical input for the AHP. This layer is essential, as areas near streams, including smaller rivers and watersheds, combined with low elevations, are significantly more prone to flooding (Oriola et al., 2016). These conditions not only heighten the risk of flooding but also amplify the likelihood of mine displacement along riverbanks and shorelines, adding complexity to both demining operations and safety planning.

4.1.1 Slope

To calculate the slope from the clipped DEM, the Spatial Analyst Tools in ArcGIS Pro were used. Specifically, the Slope tool within the Surface toolset computes the gradient for each raster cell (Esri, 2024j). The interface and settings for the tool are shown in Figure 10. Higher slope values correspond to steeper terrain, while lower values indicate flatter areas (Esri, 2024h). The output can be represented in degrees or as a percentage (percent rise), where a 45° slope corresponds to a 100% rise, and slopes approaching 90° result in an exponentially increasing percent rise (Esri, 2024h).

The planar and geodesic methods are the two primary approaches for slope calculation. The planar method calculates the slope by determining the steepest rate of change between a central cell and its neighboring cells on a flat, 2D plane using Cartesian coordinates. The slope in degrees is computed as follows:

$$\text{slope} = \arctan\left(\sqrt{\left(\frac{dz}{dx}\right)^2 + \left(\frac{dz}{dy}\right)^2}\right) \cdot 57.29578$$

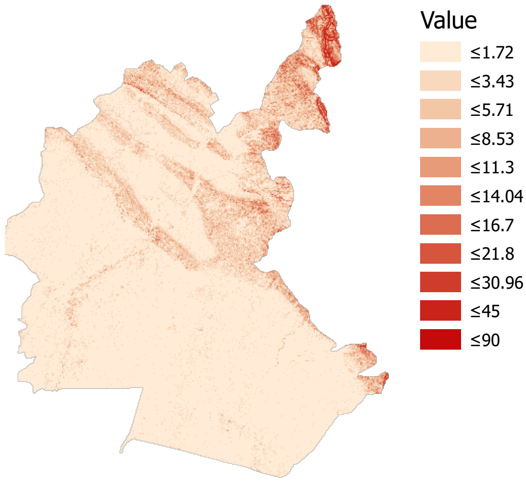

Figure 9: Visualization of the Slope Layer, illustrating variations in terrain steepness across the study area. The gradient colors represent different slope degrees, with dark red indicating slopes greater than or equal to 90 degrees and beige representing slopes less than or equal to 1 degree.

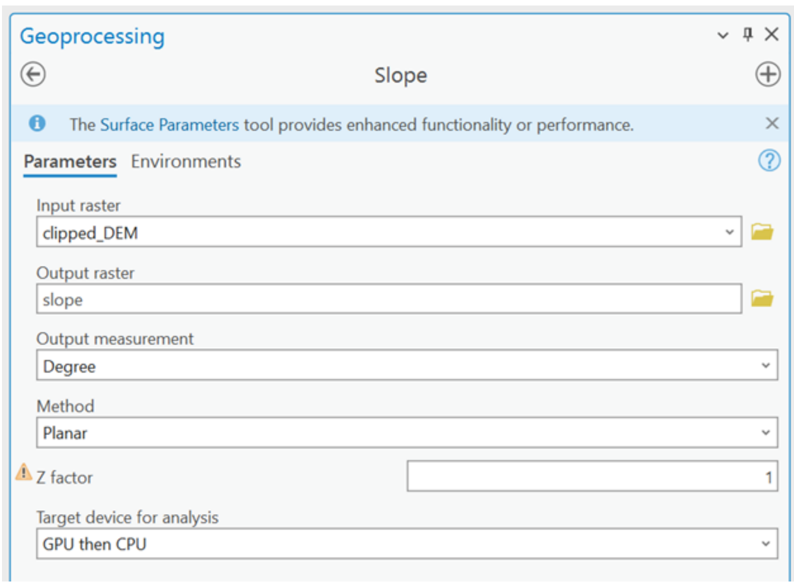

Figure 10: Slope Layer Tool in ArcGIS Pro.

a	b	c
d	e	f
g	h	i

Figure 11: Surface Scanning Window: Cell e represents the location for which the slope is calculated. The adjacent cells influence the gradient calculations in the x and y directions. Reproduced from (Esri, 2024h).

25	10	25
30	20	50
30	40	50

Figure 12: Slope Calculation Input Example: The input values for the slope calculation in this specific example assume a 5x5 cell resolution.

where 57.29578 is the conversion factor from radians to degrees (Esri, 2024h).

The partial derivatives $\frac{dz}{dx}$ and $\frac{dz}{dy}$ are determined using the following formulas:

$$\frac{dz}{dx} = \frac{(c + 2f + i) \cdot \frac{4}{\text{weight}_1} - (a + 2d + g) \cdot \frac{4}{\text{weight}_2}}{8 \cdot x_{\text{cellsize}}}$$

$$\frac{dz}{dy} = \frac{(g + 2h + i) \cdot \frac{4}{\text{weight}_3} - (a + 2b + c) \cdot \frac{4}{\text{weight}_4}}{8 \cdot y_{\text{cellsize}}}$$

where cells a through i are defined as in Figure 11, x_{cellsize} and y_{cellsize} represent the resolution of the DEM, and weight_{1-4} are the weighted counts of valid neighboring cells. Cells without valid data are assigned a weight of 0 (Esri, 2024h).

As an illustration, consider the example shown in Figure 12:

$$\frac{dz}{dx} = \frac{(25 + 2 \cdot 50 + 50) \cdot \frac{4}{(1+2\cdot1+1)} - (25 + 2 \cdot 30 + 30) \cdot \frac{4}{(1+2\cdot1+1)}}{8 \cdot 5} = 1.5$$

$$\frac{dz}{dy} = \frac{(30 + 2 \cdot 40 + 50) \cdot \frac{4}{(1+2\cdot1+1)} - (25 + 2 \cdot 10 + 25) \cdot \frac{4}{(1+2\cdot1+1)}}{8 \cdot 5} = 2.25$$

$$\text{slope} = \arctan\left(\sqrt{(1.5)^2 + (2.25)^2}\right) \cdot 57.29578 = 69.71$$

Thus, the slope for cell e is calculated to be approximately 70°.

While the geodesic method accounts for the Earth's curvature by calculating slope on a 3D ellipsoid, the planar method was selected for this case study. The terrain across the Diyala Governorate is predominantly flat, as indicated by the minimal slope variation in Figure 9. Additionally, the relatively small size of the study area ensures that neglecting the Earth's curvature will not significantly affect accuracy. Therefore, the simpler planar method provided sufficient accuracy for this analysis.

As seen in Figure 10, a critical aspect of the slope calculation is the z-factor, which corrects for differences in units between vertical (z) and horizontal (x,y) measurements.

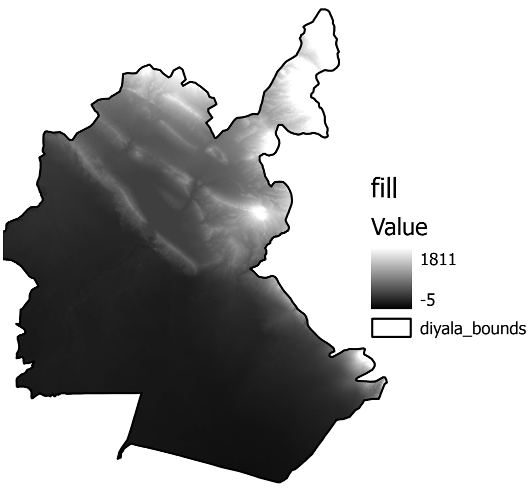

Figure 13: Fill Layer produced after applying the Fill tool.

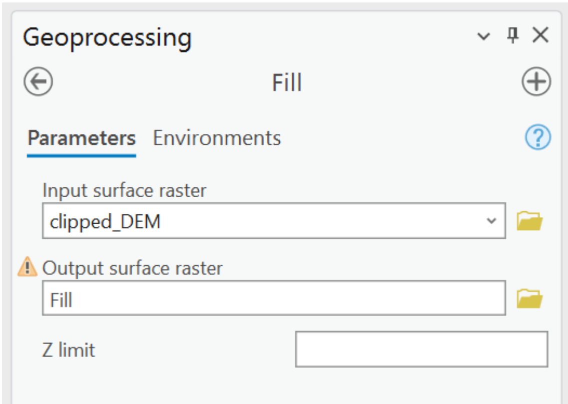

Figure 14: Fill Tool interface in ArcGIS Pro, showing key input parameters used for the analysis.

It the units are the same, the z-factor equals 1; otherwise, it must be adjusted to ensure accurate results (Esri, 2024j). The configuration settings for the Slope Tool utilized in this analysis are presented in Figure 10. The resulting slope layer, depicted in Figure 9, offers a clear visualization of terrain variations across the study area, with dark red indicating steeper slopes and beige representing flatter regions.

4.1.2 Fill

The Fill function, located in the Hydrology section of ArcGIS Pro's Spatial Analyst Tools, is essential for eliminating small depressions (sinks) in the DEM that could interfere with accurate flow modeling in subsequent analyses. This tool ensures that surface water can flow continuously without getting trapped in low-lying cells. The Fill tool applies a series of geoprocessing functions, including Focal Flow, Flow Direction, Sink, Watershed, and Zonal Fill, to detect and remove these sinks (Esri, 2024f). This process is depicted in Figure 15, while Figure 14 illustrates the tool's interface and the parameters used in this analysis.

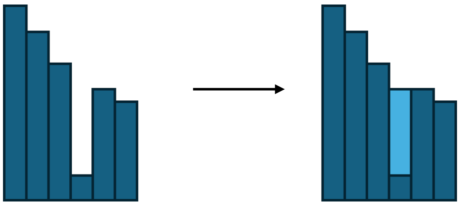

Figure 15: Visualization of the Fill process, depicting how depressions in the DEM are corrected by filling them up to their pour point elevation. Reproduced from (Esri, 2024f).

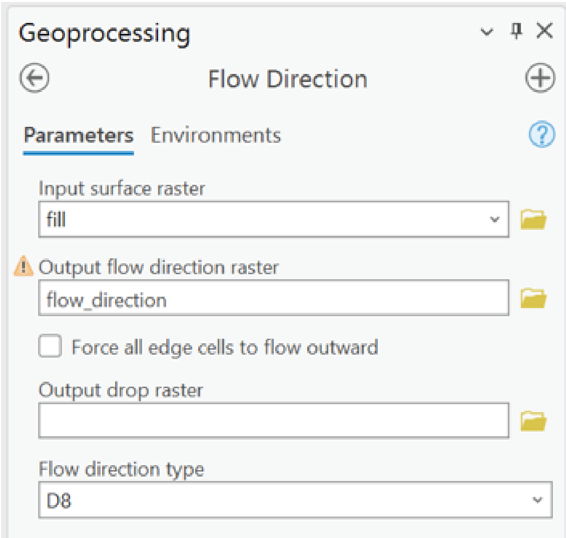

Figure 16: Flow Direction Tool and its input parameters, including the Fill layer as the key input for generating the flow direction raster.

A sink is defined as a cell or group of cells in the DEM that lack a defined drainage direction, typically because they are surrounded by higher-elevation cells. This disrupts flow accumulation, which is crucial for hydrological analysis. The Fill tool resolves this issue by identifying each sink's pour point — the lowest boundary where water would naturally flow if the sink were filled. Once identified, the tool raises the elevation of the sink to match that of the pour point, effectively creating a continuous surface (Esri, 2024c).

A critical parameter in the fill process is the z-limit, which defines the maximum allowable elevation difference between the lowest point of a sink and its pour point that can be filled. For this analysis, no z-limit was applied, ensuring that all depressions were filled regardless of depth (Esri, 2024c). However, in other cases, setting an appropriate z-limit is essential for accurately modeling surface flow and should be done by experts familiar with the terrain and hydrology of the study area. An incorrect z-limit can result in either underfilling, leaving substantial depressions that disrupt drainage networks, or overfilling, which may distort the natural landscape and lead to an inaccurate depiction of surface flow. The modified terrain, with all depressions filled to their pour point elevations, is visualized in the Fill Layer presented in Figure 13.

4.1.3 Flow Direction

The Flow Direction tool in ArcGIS Pro is essential for modeling water movement across a landscape. It calculates the flow direction for each cell in a raster by identifying the steepest downslope neighbor, utilizing methods such as the widely used *Deterministic*

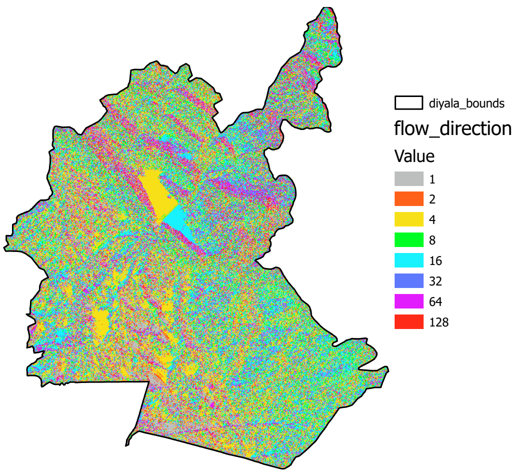

Figure 17: Flow Direction Layer, visualizing the direction of water flow across the study area based on the steepest slope for each raster cell.

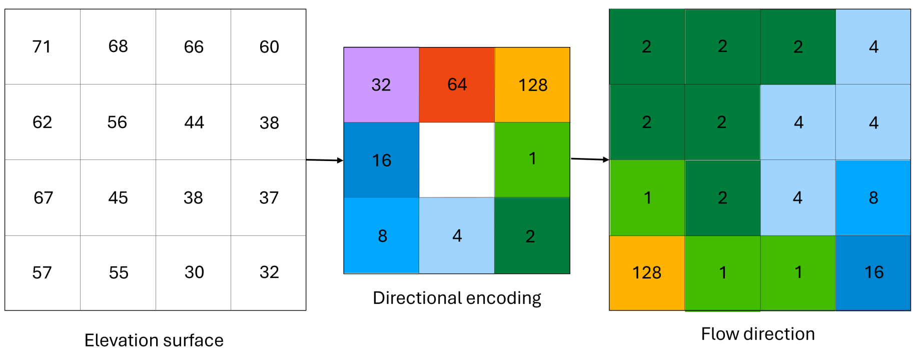

Figure 18: Example of Flow Direction Raster Generation

64	128	1
32		2
16	8	4

Figure 19: Original directional encoding of flow directions as proposed by Jenson et al. Reproduced from (Jenson et al., 1988).

32	64	128
16		1
8	4	2

Figure 20: Directional encoding used in ArcGIS Pro's Flow Direction tool, adapted from the original encoding. Reproduced from (Esri, 2024g).

8-neighbor (D8), or the more advanced *Multiple Flow Direction (MFD)* and *D-Infinity (DINF)* techniques (Esri, 2024e). For this analysis, the D8 method was chosen due to its simplicity and efficiency, assigning flow directions based on the steepest descent to neighboring cells. The input for the Flow Direction tool, as shown in Figure 16, is a depressionless DEM, also known as a fill layer. This ensures that water flows correctly across the landscape, without being obstructed by artificial depressions or sinks (Jenson et al., 1988). The output of this tool is the flow direction layer seen in Figure 17, which visually represents the direction of water flow for each cell. The various colors in the layer correspond to the different flow directions identified by the tool.

Using the D8 method, flow direction is determined by calculating the steepest descent from each cell to its eight neighboring cells. Each direction is assigned a specific value that reflects this steepest drop. The original directional encoding proposed by Jenson et al. is displayed in Figure 19, while the encoding used in ArcGIS Pro is shown in Figure 20. The primary difference between these encodings is that the ArcGIS Pro version shifts the values one step to the right compared to the original.

To clarify this process, Figure 18 presents an illustrative example. The leftmost raster represents the depressionless DEM used as the input. The second raster is the directional encoding from Figure 19, and the third raster shows the resulting layer after running the tool.

The flow direction for each raster cell is determined by one of the following four conditions:

1. Single-Cell Depression: When the central cell is surrounded by higher elevation cells, forming an isolated depression, a negative flow direction value is assigned. Though the Fill tool typically eliminates such depressions (as described in Section 4.1.2), this condition is still considered for completeness (Jenson et al., 1988).

2. Steepest Descent: The most common case, where one neighboring cell exhibits the steepest descent relative to the center cell. Flow direction is assigned to this neighbor based on the steepest distance-weighted drop, calculated using:

$$\text{distance-weighted drop} = \frac{\text{z-value change}}{\text{distance}}$$

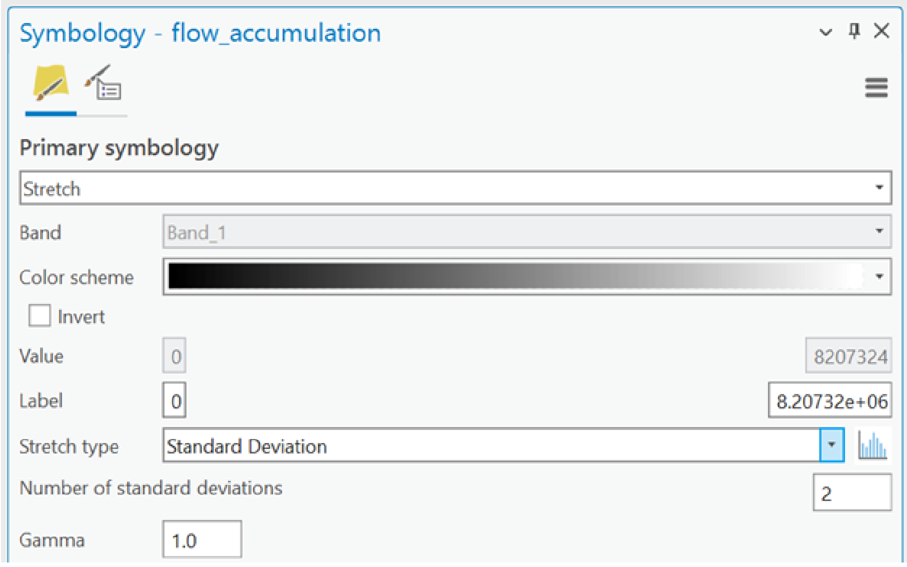

Figure 21: Symbology Stretch Type: Standard Deviation for Enhanced Display.

Figure 22: Zoomed-In View of the Flow Accumulation Layer.

where the z-value change represents the elevation difference between the center cell and its neighboring cell. Diagonal neighbors have a distance of $\sqrt{2}$, while directly adjacent neighbors are at a distance of 1 (Jenson et al., 1988).

3. Equal Descent: When multiple neighboring cells have equal distance-weighted drops, a logical lookup operation determines the flow direction. For example, if adjacent cells along one edge have equal drops, the center cell among them is typically assigned the flow direction. If cells on opposite sides have equal drops, one direction is chosen arbitrarily (Jenson et al., 1988).

4. Flat Area: If all neighboring cells have equal elevations, indicating a flat region, an iterative process assigns flow directions by connecting flat cells to those with defined flow directions, avoiding loops. This process continues until all cells in the flat area have assigned flow directions (Jenson et al., 1988).

4.1.4 Flow Accumulation

The Flow Accumulation tool generates a raster representing the total accumulated flow to each cell by summing the contributions from all upstream cells. Optionally, a weight raster, as seen in Figure 23 showing the interface of the tool, can be applied to influence

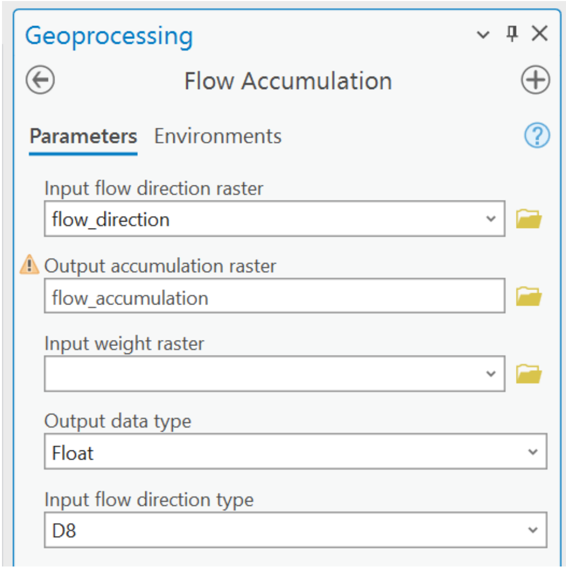

Figure 23: Interface of the Flow Direction Tool and Its Inputs

this accumulation. The tool supports three different flow modeling algorithms: D8, MFD and DINF, each of which affects how flow is distributed and accumulated across the cells. It is essential to use the same method that was used to generate the input flow direction raster, as inconsistencies may cause issues such as loops in the flow direction, leading to an infinite loop during the accumulation process. Typically, the flow direction raster is created using the Flow Direction tool to ensure compatibility (Esri, 2024d).

Cells with undefined flow direction will only receive flow from upstream cells but will not contribute to downstream accumulation. For instance, in a D8 flow direction raster, a cell is considered undefined if its value does not match one of the standard direction codes shown in Figure 20. The accumulation is computed by counting the number of upstream cells contributing flow to each downslope cell in the output raster. If no weight raster is specified, as is the case in this analysis, the tool assumes a default weight of 1, meaning that each upstream cell contributes equally to the flow accumulation (Esri, 2024d).

For better visualization, the Standard Deviation stretch type can be applied to the flow accumulation layer, as illustrated in Figure 21 displaying the Symbology interface. This technique enhances the contrast between areas of high and low accumulation, making key features such as stream channels and ridges more discernible. A sample of this visualization, randomly zoomed in, is shown in Figure 22.

Areas with high flow accumulation typically correspond to concentrated flow paths, such as potential stream channels. Conversely, areas with zero flow accumulation usually indicate local topographic highs, such as ridges (Esri, 2024d). As a sanity check, it is crucial to confirm that high accumulation values correspond with known or visible water channels in the landscape.

4.1.5 Streams

The flow accumulation results are essential for delineating stream networks, achieved by applying a threshold to identify cells with accumulated flow. For this analysis, cells with a flow accumulation exceeding 1% of the maximum value were designated as part of the stream network (Zhang et al., 2021). With the dimensionless maximum flow accumulation value being 8,207,320, this threshold is set at 82,073.2. As shown in Table 4, cells with a flow accumulation above this threshold are classified as part of the stream network (assigned a value of 1), while others are marked as NoData.

Table 4: Stream Network Extraction Classification

Flow Accumulation Range	Reclassified Value
0 - 82073.2	NoData
82073.2 - 8207320	1

Figure 24: Zoomed-In Visualization of the Stream Layer

As depicted in Figure 24, this threshold effectively captures the primary streams within the study area. Adjusting this threshold could reveal finer details within the stream network, but may also result in overclassification, where areas that are not streams are mistakenly included in the network. Therefore, determining the optimal threshold requires balancing detail with accuracy, often involving iterative adjustments (Zhang et al., 2021).

To generate the stream network raster, several methods can be employed:

- Con Tool (Spatial Analyst): This method involves inputting the flow accumulation layer as the conditional raster, setting the expression to select values greater than 82,073.2, and assigning an output value of 1 for stream areas.
- Set Null Tool (Spatial Analyst): This method uses the flow accumulation layer as the input conditional raster, with an expression to exclude values less than or equal to 82,073.2, setting the output value to 1 for streams
- Reclassify Tool (Spatial Analyst): In this method, the flow accumulation layer is input, and a manual classification is applied to divide the data into two classes — streams with a value of 1 and non-stream areas with NoData.

In this case study, the Reclassify method was chosen for generating the stream network. The settings of the Reclassify Tool can be seen in Figure 25. This method, and its relevance to the reclassification process, will be discussed in greater detail in Section 4.2.

4.1.6 Stream Proximity

The final layer generated in this analysis is the Stream Proximity layer. This layer is crucial for determining the proximity of each area to the nearest water body, a key factor in flood hazard mapping. Areas closer to water bodies are generally more susceptible to flooding, making proximity a vital criterion in assessing flood risk (Yodying et al., 2019).

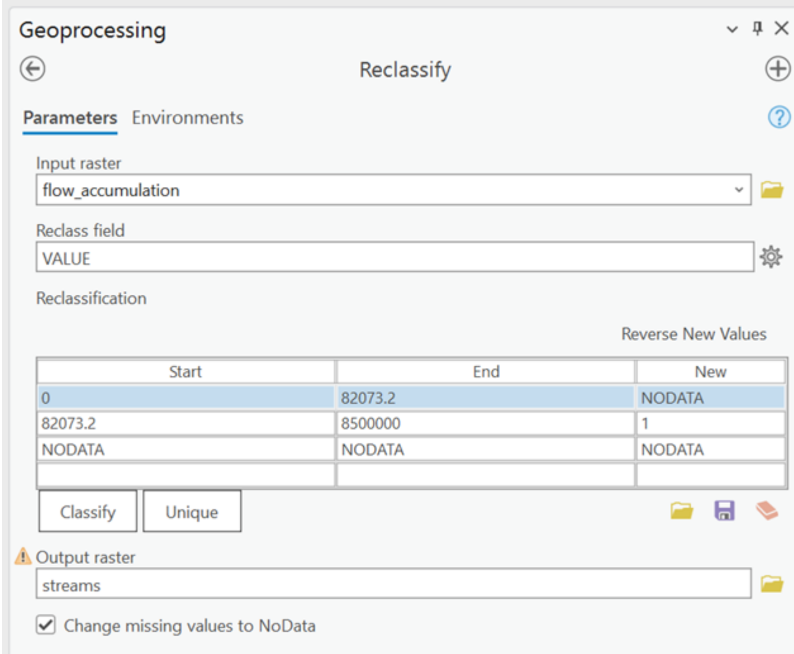

Figure 25: Obtaining Streams from Flow Accumulation

The primary goal of modifying the DEM in earlier steps was to accurately capture this proximity.

To create this layer, the Euclidean Distance tool from the Spatial Analyst toolbox depicted in Figure 26 was employed. This tool calculates the straight-line distance from each cell in a raster to the nearest stream, providing a direct measure of proximity to the stream network (Esri, 2024b).

In this analysis, the input for the Euclidean Distance tool was a raster where source cells corresponded to valid values from the previously created Stream layer. Cells designated as NoData were excluded, while cells with a value of 0 were considered valid and included in the proximity calculation. The Euclidean Distance algorithm determines the shortest path between cells by calculating the hypotenuse of a right triangle formed by the horizontal and vertical distances. The resulting output raster assigns each cell a value representing its distance from the nearest stream. If a cell is equidistant from multiple streams, the tool assigns it to the first stream encountered during the scan (Esri, 2024b)

It is important to note that the Euclidean Distance tool assumes a straight-line path, which may not always reflect reality due to natural barriers such as ridges or steep slopes. In this case study, the simpler planar distance method was applied, which is appropriate for the study area given its topographic characteristics, as discussed in Section 4.1.1. The resulting Stream Proximity layer, shown in Figure 27, effectively highlights areas in the study region, with green indicating proximity to streams.

4.2 Reclassification

Reclassification is a crucial step in simplifying continuous raster data into discrete, interpretable classes, enabling more straightforward analysis and comparison of factors contributing to flood risk. This process involves categorizing continuous raster values into predefined classes, making it easier to assess and compare the varying levels of flood susceptibility across the study area. For this case study, a five-class classification scheme was employed, following the methodology proposed by Roopnarine et al. The classes are: Very Low, Low, Moderate, High, and Very High, as detailed in Table 5.

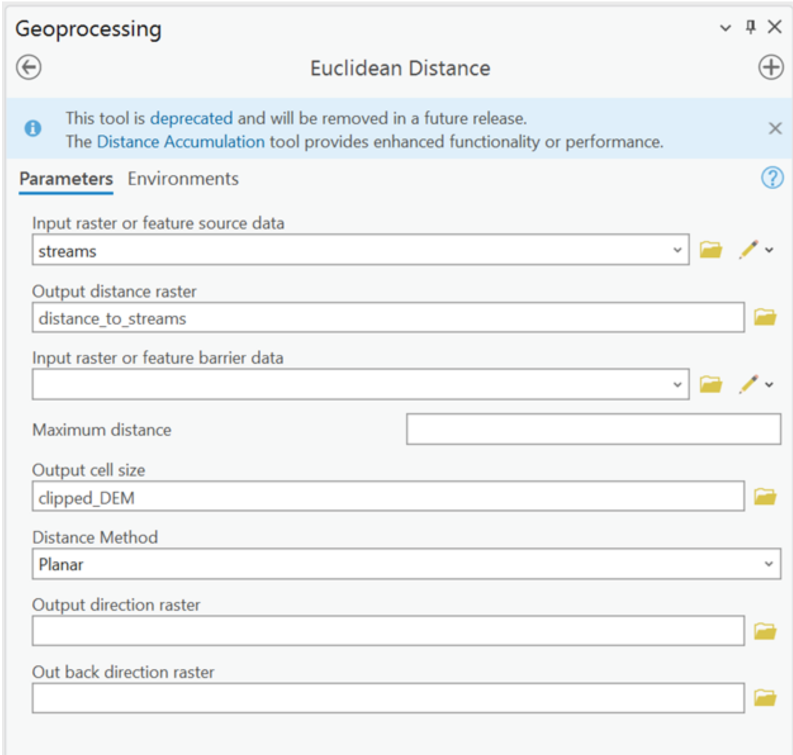

Figure 26: Euclidean Distance Tool Interface

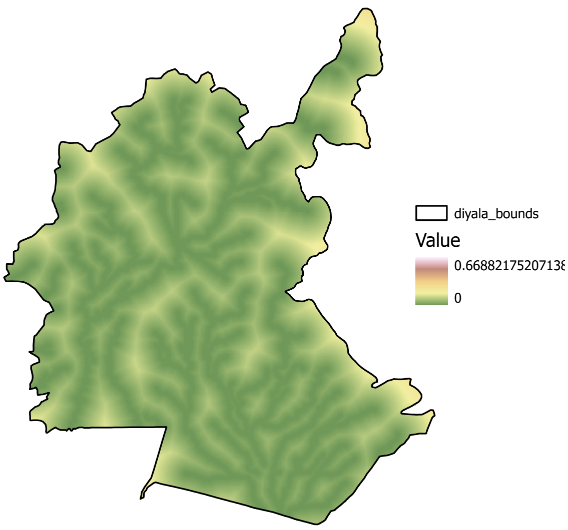

Figure 27: Proximity to Streams Layer Illustrating Areas Based on Distance to Water Sources. Green areas indicate regions in close proximity to streams.

Table 5: Five-Class Flood Susceptibility Classification Scheme

1	**2**	**3**	**4**	**5**
Very Low	Low	Moderate	High	Very High

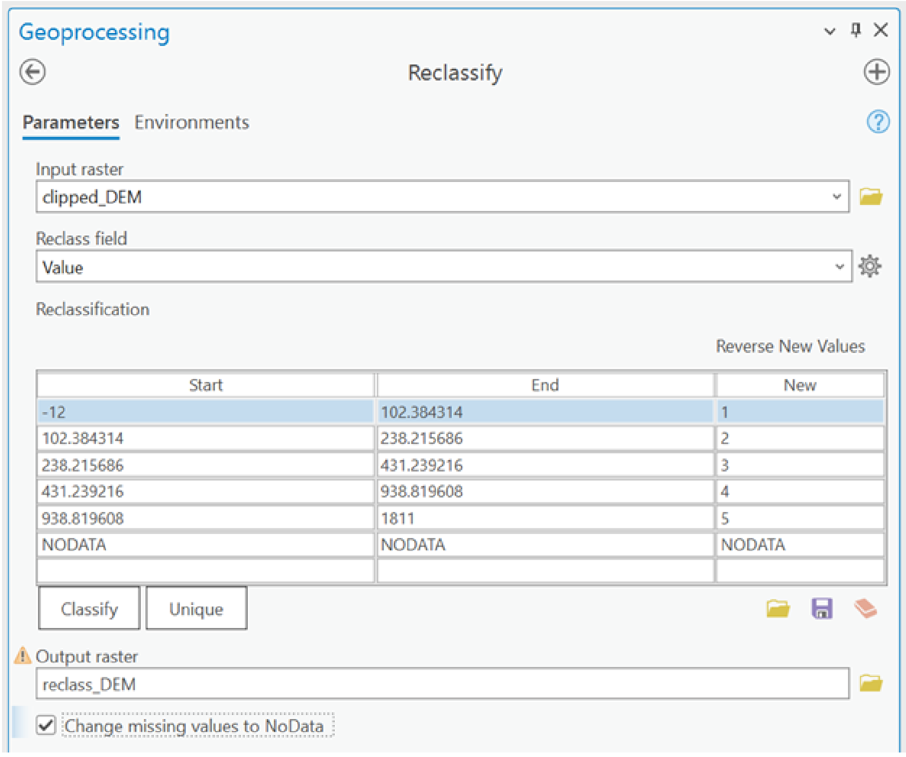

Figure 28: Reclassification of DEM

Table 6: Reclassified Land Cover Types Based on Flood Susceptibility

Name	**Class**	**Class According to Classification Scheme**
Water	1	5
Trees	2	2
Flooded Vegetation	4	5
Crops	5	4
Built Area	7	4
Bare Ground	8	3
Rangeland	11	3

Several methods are available for reclassification, such as Natural Breaks (Jenks), Quantile, Equal Interval, Defined Interval, Geometric Interval, and Standard Deviation (Esri, 2024a). For this analysis, the Natural Breaks (Jenks) method was primarily used alongside Manual Interval reclassification. Natural Breaks (Jenks) classification is a data-driven method used to optimize the division of a dataset into distinct classes based on inherent patterns within the data itself. The algorithm identifies class boundaries in such a way that it minimizes the variance within each class while maximizing the variance between classes. This method ensures that similar data values are grouped together into distinct classes, maximizing the differences between them to reflect meaningful patterns within the dataset (Esri, 2024a). To apply the Natural Breaks (Jenks) classification method, navigate to the "Classify" option within the Reclassify Tool interface shown in Figure 28. From there, select the appropriate method and specify the desired number of classes.

For the land cover layer, a different approach was necessary. The "Unique" option in the Reclassify tool was utilized, allowing for the manual assignment of flood susceptibility classes based on the specific characteristics of each land cover type. Table 6 summarizes the reclassified land cover types according to their flood susceptibility. .

The classification of each land cover type is justified as follows (Roopnarine et al., 2022):

- Water bodies are naturally highly susceptible to flooding due to their existing saturation, thus they are classified as having very high susceptibility.
- Dense tree cover can affect water flow and absorb significant amounts of water, making it less susceptible to flooding compared to bare ground or built areas. It is therefore classified as low susceptibility.
- Flooded vegetation indicates areas prone to frequent flooding, warranting a classification of very high susceptibility.
- Agricultural areas, particularly those with poor drainage, can be prone to flooding, although they are generally less susceptible than water bodies or flooded vegetation. These are classified as high susceptibility.
- Built areas tend to have impervious surfaces, increasing runoff and the risk of flooding, justifying a classification of high susceptibility.
- Bare ground, while susceptible to erosion, is classified as having moderate susceptibility to flooding, unless it is situated in a particularly low-lying area.
- Rangeland, which typically consists of open, grassy areas, can absorb water but may become saturated depending on the amount of rainfall. It is classified as having moderate susceptibility.

4.3 Weighting Factors

In flood hazard mapping, determining appropriate weighting factors is essential to accurately assess the relative importance of various criteria. The objective is to derive a set of weights that consistently and logically reflect each criterion's contribution to flood risk. This method, successfully applied in case studies such as those by Shuaibu et al., Yodying et al., and Ekeanyanwu et al., involves a structured process that ensures each criterion is prioritized appropriately, leading to a more precise and meaningful analysis.

The AHP begins with a clear definition of the objective: to produce a flood hazard map that integrates multiple criteria. Selecting relevant criteria is crucial, as these will form the foundation of the analysis. This case study uses the same criteria established by Ekeanyanwu et al. Once the criteria are established, they are systematically compared using a pairwise comparison matrix:

$$\mathbf{A_1} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

This matrix, denoted as A_1, enables the evaluation of each criterion against the others, capturing the relative importance in contributing to flood risk. The matrix values are determined based on expert judgement or empirical data to ensure consistent and logical comparisons.

Next, λ_{max} values are calculated for each criterion using the geometric mean of the elements in each row:

$$\lambda_{max_j} = \prod_{j=1}^{n} (a_{jn})^{1/n}$$

Table 7: RI for Different Number of Flood Hazard Criteria n. Reproduced from (Yodying et al., 2019)

n	1	2	3	4	5	6	7	8	9	10
RI	0	0	0.58	0.90	1.12	1.24	1.32	1.41	1.45	1.49

These values provide a quantitative measure of each criterion's overall priority. The final weights (w_i) are then derived by normalizing the λ_{max} values:

$$w_i = \frac{\lambda_{max_i}}{\sum_{j=1}^{n} \lambda_{max_j}}$$

These weights represent the proportionate significance of each factor in the flood hazard mapping process, ensuring that the final map appropriately reflects the varying impacts of each criterion.

To validate the consistency of the pairwise comparisons, a *Consistency Ratio (CR)* is calculated using the formula:

$$\mathrm{CR} = \frac{\mathrm{CI}}{\mathrm{RI}} < 0.1$$

where the *Consistency Index (CI)* is defined as:

$$CI = \frac{\lambda - n}{n-1}$$

Here, n is the total number of criteria, and the *Random Index (RI)* depends on this number, as detailed in Table 7. To obtain λ, three matrices must be calculated:

$$\mathbf{A_2} = \begin{pmatrix} w_1 \\ \vdots \\ w_i \end{pmatrix}, \ \mathbf{A_3} = \mathbf{A_1} \times \mathbf{A_2}, \ \mathbf{A_4} = \frac{\mathbf{A_3}}{\mathbf{A_2}}$$

The value of λ is then calculated as the average of the sum of the entries in matrix A_4:

$$\lambda = \frac{1}{m} \sum_{i=1}^{m} (\mathbf{A_4})_i$$

where m is the number of rows in A_4 and $(A_4)_i$ represents the ith entry in the matrix. A CR value below 0.1 indicates that the comparisons are consistent, confirming that the derived weights are reliable.

Once established and validated, these weights can be used to integrate the various criteria into a comprehensive FHM. This methodical approach ensures that the most influential factors receive appropriate emphasis, resulting in a more accurate and reliable flood risk assessment for the study area. The following sections will detail the specific calculations used in this case study, demonstrating the practical application of this process.

4.3.1 Weight Calculation based on Shuaibu et al.

In the first set of FHMs, the criteria considered include elevation, slope, land cover, precipitation, and stream proximity (Ekeanyanwu et al., 2022). The pair-wise comparison matrix values, derived from the study by Shuaibu et al., are presented in Table 8. The resulting λ_{max} values for each criterion are shown Table 9.

Table 8: AHP Pair-Wise Comparison Matrix (Based on Shuaibu et al.)

	Elevation	**Slope**	**Land Cover**	**Precipitation**	**Stream Proximity**
Elevation	1	5	3	3	3
Slope	1/5	1	1/3	1/3	1/3
Land Cover	1/3	3	1	1/3	1/3
Precipitation	1/3	3	3	1	1/3
Stream Proximity	1/3	3	3	3	1

Table 9: λ_{max} Values for Each Criterion (Based on Shuaibu et al.)

Criterion	**Expression**
Elevation	$\lambda_{max_1} = (1 \cdot 5 \cdot 3 \cdot 3 \cdot 3)^{1/5} = 2.66727$
Slope	$\lambda_{max_2} = (1/5 \cdot 1 \cdot 1/3 \cdot 1/3 \cdot 1/3)^{1/5} = 0.37492$
Land cover	$\lambda_{max_3} = (1/3 \cdot 3 \cdot 1 \cdot 1/3 \cdot 1/3)^{1/5} = 0.64439$
Precipitation	$\lambda_{max_4} = (1/3 \cdot 3 \cdot 3 \cdot 1 \cdot 1/3)^{1/5} = 1$
Stream Proximity	$\lambda_{max_5} = (1/3 \cdot 3 \cdot 3 \cdot 3 \cdot 1)^{1/5} = 1.55185$

The final weights for each criterion, summarized in Table 10, are calculated as follows:

$$\mathbf{A_2} = w = \begin{pmatrix} 2.66727/6.23843 \\ 0.37492/6.23843 \\ 0.64439/6.23843 \\ 1/6.23843 \\ 1.55185/6.23843 \end{pmatrix} = \begin{pmatrix} 0.42755 \\ 0.0601 \\ 0.10329 \\ 0.1603 \\ 0.24876 \end{pmatrix}$$

Finally, the CR is verified to ensure reliability of the weights:

$$\mathbf{A_1} = \begin{pmatrix} 1 & 5 & 3 & 3 & 3 \\ 1/5 & 1 & 1/3 & 1/3 & 1/3 \\ 1/3 & 3 & 1 & 1/3 & 1/3 \\ 1/3 & 3 & 3 & 1 & 1/3 \\ 1/3 & 3 & 3 & 3 & 1 \end{pmatrix}, \ \mathbf{A_3} = \begin{pmatrix} 2.2651 \\ 0.31639 \\ 0.56246 \\ 0.87591 \\ 1.36235 \end{pmatrix}, \ \mathbf{A_4} = \begin{pmatrix} 5.29786 \\ 5.26444 \\ 5.44544 \\ 5.46417 \\ 5.47656 \end{pmatrix}$$

$$\lambda = 5.389694, \ CI = \frac{5.389694 - 5}{5 - 1} = 0.0974235, \ CR = \frac{0.0974235}{1.12} = 0.087 < 0.1$$

Here, RI is 1.12, as shown in Table 7. The CR value below 0.1 confirms the consistency and reliability of the calculated weights.

Table 10: Percentage Weighting of Flood Hazard Criteria based on Shuaibu et al.

Criterion	**Percentage Weight**
Elevation	43%
Slope	6%
Land cover	10%
Precipitation	16%
Stream Proximity	25%

Table 11: AHP Pair-Wise Comparison Matrix (Based on Yodying et al.)

	Elevation	**Slope**	**Land Cover**	**Precipitation**	**Stream Proximity**
Elevation	1	3	2	2	2
Slope	$1/3$	1	$1/2$	$1/2$	$1/2$
Land Cover	$1/2$	2	1	$1/2$	$1/2$
Precipitation	$1/2$	2	2	1	$1/2$
Stream Proximity	$1/2$	2	2	2	1

Table 12: λ_{max} Values for Each Criterion (Based on Yodying et al.)

Criterion	**Expression**
Elevation	$\lambda_{max_1} = (1 \cdot 3 \cdot 2 \cdot 2 \cdot 2)^{1/5} = 1.88818$
Slope	$\lambda_{max_2} = (1/3 \cdot 1 \cdot 1/2 \cdot 1/2 \cdot 1/2)^{1/5} = 0.52961$
Land cover	$\lambda_{max_3} = (1/2 \cdot 2 \cdot 1 \cdot 1/2 \cdot 1/2)^{1/5} = 0.75786$
Precipitation	$\lambda_{max_4} = (1/2 \cdot 2 \cdot 2 \cdot 1 \cdot 1/2)^{1/5} = 1$
Stream Proximity	$\lambda_{max_5} = (1/2 \cdot 2 \cdot 2 \cdot 2 \cdot 1)^{1/5} = 1.31951$

4.3.2 Weight Calculation Based on Yodying et al.

In this second approach to generating FHMs, the criteria considered remain the same: elevation, slope, land cover, precipitation, stream proximity (Ekeanyanwu et al., 2022). However, the weights assigned to these criteria are derived from the methodology used by Yodying et al. The pair-wise comparison matrix values, reflecting the relative importance of each criterion according to their research, is provided in Table 11. The resulting λ_{max} values for each criterion are shown in Table 12.

Following the calculation of λ_{max} values, the final weights assigned to each criterion are derived. The weights are summarized in Table 13, and calculated as follows:

$$\mathbf{A_2} = w = \begin{pmatrix} 1.88818/5.49516 \\ 0.52961/5.49516 \\ 0.75786/5.49516 \\ 1/5.49516 \\ 1.31951/5.49516 \end{pmatrix} = \begin{pmatrix} 0.34361 \\ 0.09638 \\ 0.13791 \\ 0.18198 \\ 0.24012 \end{pmatrix}$$

To validate the reliability of these weights, the CR is calculated using the following

Table 13: Percentage Weighting of Flood Hazard Criteria (Based on Yodying et al.)

Criterion	**Percentage Weight**
Elevation	34%
Slope	10%
Land cover	14%
Precipitation	18%
Stream Proximity	24%

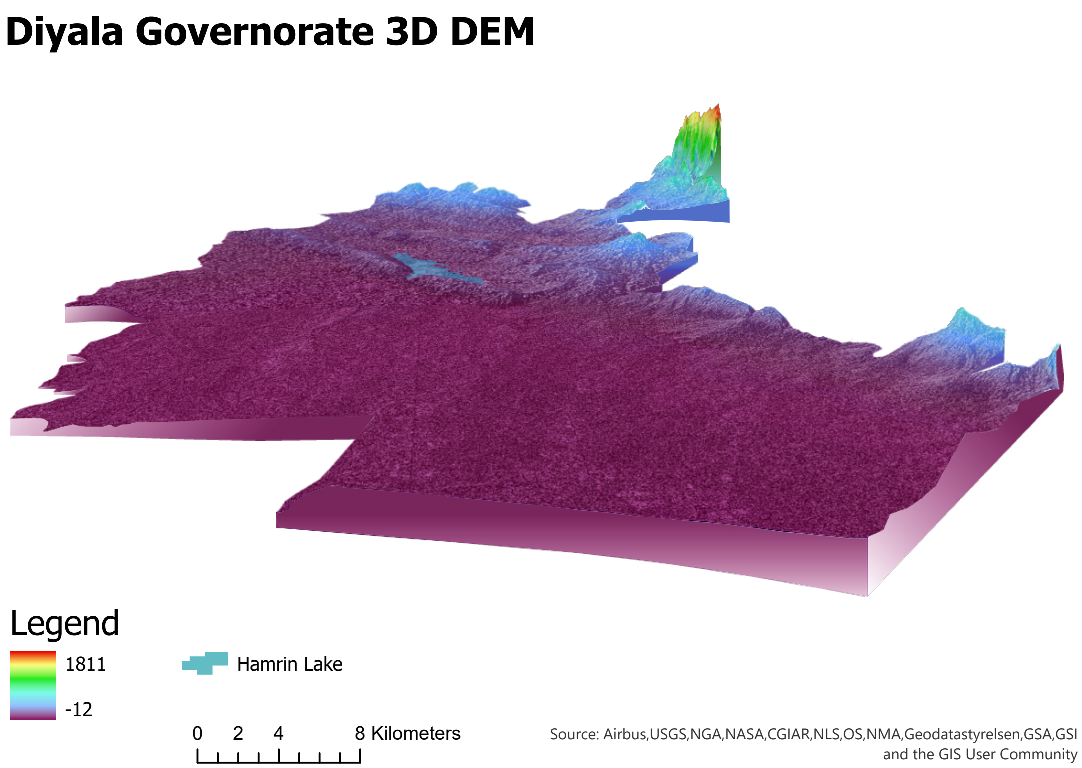

Figure 29: Three-Dimensional Digital Elevation Model of the Study Area, with Hamrin Lake visible in blue.

matrices:

$$\mathbf{A_1} = \begin{pmatrix} 1 & 3 & 2 & 2 & 2 \\ 1/3 & 1 & 1/2 & 1/2 & 1/2 \\ 1/2 & 2 & 1 & 1/2 & 1/2 \\ 1/2 & 2 & 2 & 1 & 1/2 \\ 1/2 & 2 & 2 & 2 & 1 \end{pmatrix}, \; \mathbf{A_3} = \begin{pmatrix} 1.75277 \\ 0.49092 \\ 0.71352 \\ 0.94243 \\ 1.24447 \end{pmatrix}, \; \mathbf{A_4} = \begin{pmatrix} 5.10104 \\ 5.09360 \\ 5.17384 \\ 5.17872 \\ 5.18270 \end{pmatrix}$$

$$\lambda = 5.14598, \; CI = \frac{5.14598 - 5}{5 - 1} = 0.036495, \; CR = \frac{0.036495}{1.12} = 0.0326 < 0.1$$

Given that the CR is less than 0.1, the pair-wise comparisons are consistent, confirming that the weights derived are reliable.

5 Results and Discussion

This section presents the result of the flood hazard mapping analysis and discusses the implications of using different weighting criteria. The case study generated several FHMs based on historical and predictive data. The results highlight the impact of these variations in weighting on the identification and classification of flood-prone areas. The subsections below provide a detailed comparison of the flood hazard maps derived from the different sets of weights, as well as a focused analysis of specific hazardous areas, such as those around the mines.

5.1 Flood Hazard Maps using Weights Based on Shuaibu et al.

The first set of FHMs was generated using the comparison matrix values proposed by Shuaibu et al. As detailed in Table 10, elevation was assigned the highest weight at 43%,

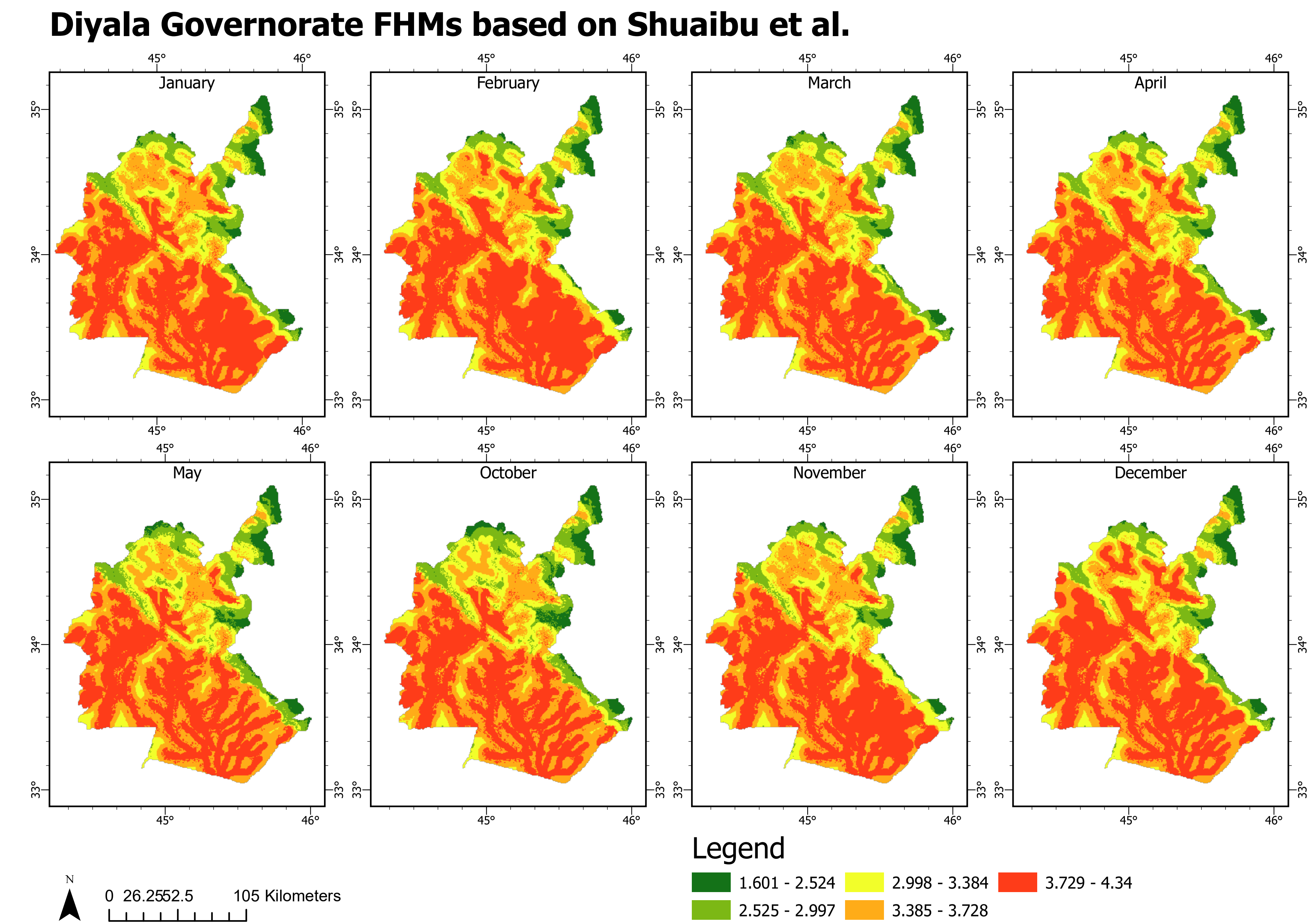

Figure 30: Flood Hazard Maps (FHMs) based on Shuaibu et al. These maps represent flood hazard levels from October through May, classified into five risk categories ranging from very low (green) to very high (red).

Table 14: Percentage Difference in Weighting Criteria between Flood Hazard Maps (FHMs)

Criterion	**Difference in %**
Elevation	$\|43-34\| = \Delta 9$
Slope	$\|6-10\| = \Delta 4$
Land cover	$\|10-14\| = \Delta 4$
Precipitation	$\|16-18\| = \Delta 2$
Stream Proximity	$\|25-24\| = \Delta 1$

making it the most influential factor in the flood hazard assessment. This was followed by stream proximity at 25%, precipitation at 16%, land cover at 10% and slope at 6%. Table 10 highlights how the dominance of elevation in this model emphasizes the lower risk in higher-altitude areas in the northeast.

Figure 30 presents the FHMs, which have been classified into five distinct flood hazard risk categories: very low (green), low (light green), moderate (yellow), high (orange), and very high (red). These maps cover the flood hazard assessments from October through May, capturing the critical months when flooding is most likely to occur due to seasonal variations in precipitation and water flow.

The FHMs in Figure 30 identify a significant portion of the study area as high to very high-risk zones, particularly concentrated along the river valleys in the western and southern regions. These areas, as indicated by the orange to red zones on the maps, are at the greatest risk of flooding. This pattern is consistent with the topographical features depicted in the 3D DEM shown in Figure 29. The DEM illustrates the study area's elevation, ranging from 1811 meters above sea level (highlighted in red) to -12 meters below sea level (indicated in purple). The prominent Hamrin Lake is visible in pale blue. The 3D DEM underscores how the low elevation regions in the West and South are more susceptible to flood events.

A notable pattern observed in these FHMs is the expansion of flood zones within the southern river system during the months of January, February, and November. During these months, the areas of high-risk (represented in red) expand considerably, particularly in the lower-elevation region to the west of Diyala's mountainous border visible in Figure 29. This seasonal shift highlights the dynamic nature of flood risks, driven by precipitation peaks during these months.

5.2 Flood Hazard Maps based on Yodying et al.

The second set of flood hazard maps was developed using the comparison matrix values from Yodying et al. As detailed in Table 13, elevation continues to hold the highest weight at 34%, although this is slightly lower compared to the first model. Stream proximity and precipitation follow with 24% and 18% respectively. Land cover and slope were given more importance in this model, weighted at 14% and 10%.

The differences in these weighting factors, though relatively small, result in significant variations in the classification of flood hazard areas. As shown in Table 14, elevation, slope, and land cover weights differ by 9%, 4%, and 4%, respectively, compared to the previous model. These differences, while modest, lead to a more selective identification of high to very high-risk areas in the FHMs.

Figure 31 illustrates the FHMs generated using the weights detailed in Table 13, covering the flood hazard assessments from October through May and classified into the same five

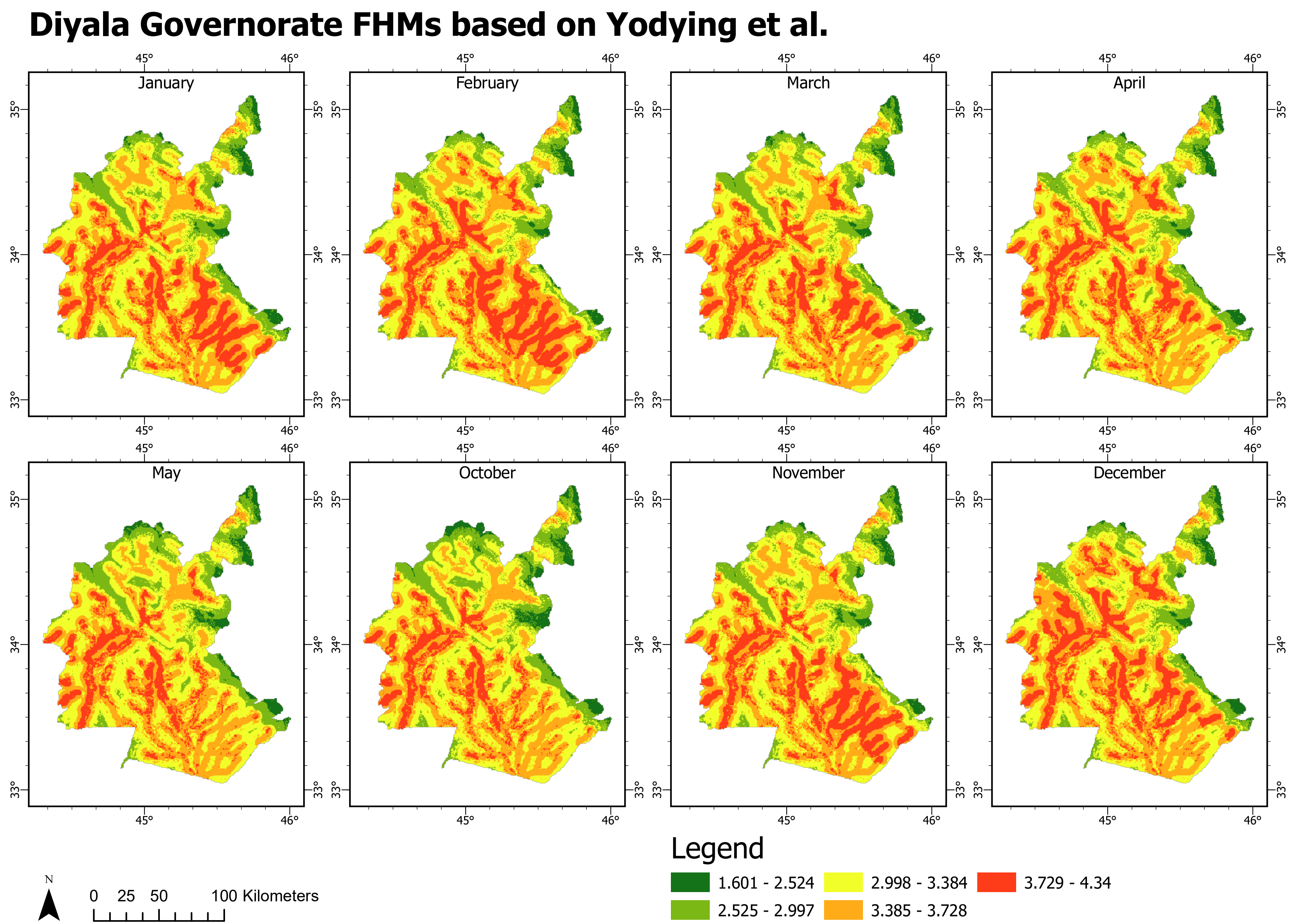

Figure 31: Flood Hazard Maps (FHMs) based on Yodying et al. These maps are classified into five risk categories, ranging from very low (green) to very high (red), and cover the flood hazard assessments from October through May.

risk categories as before. Overall, these FHMs are more selective in identifying high to very-high risk areas, as indicated by the reduced presence of red and orange zones. The highest-risk areas (in red) remain concentrated near riverbanks.

Notably, the flood hazard in the southern river system remains at its worst during January, February, and November, with the most extensive high-risk zones observed during these months. Conversely, the risk is at its lowest in May and October. The increased selectivity of this model makes it particularly useful for planning demining efforts, where precision in identifying high-risk flood areas is crucial to focus efforts on the most affected regions. Due to the effectiveness of this weighting scheme, the FHMs based on predictive precipitation data, discussed in the following section, were also generated using this weight set.

5.3 Predictive Flood Hazard Maps

The third set of FHMs is based on predictive precipitation data from the CMIP6 scenarios, using the HadGEM3-GC31-LL model. These predictions, which simulate different greenhouse gas emission trajectories (SS126, SS245, and SS585), provide insight into how future climate changes could influence flood hazards in the study area. The weights for these maps are consistent with those used in Section 5.2, allowing for a direct comparison between historical and future flood risks under varying climate conditions.

Figure 32, Figure 33, and Figure 34 present the predictive FHMs under three different emission scenarios: SS126 (low emissions), SS245 (moderate emissions), and SS5835 (high emissions), respectively. The risk categories remain the same, ranging from very low (green) to very high (red). Figure 32 illustrates the predicted flood hazards under the SS126 low-emissions scenario, representing the ”best case” trajectory in terms of climate change mitigation. Even under this optimistic scenario, there is a noticeable increase in high-risk flood zones compared to the historical FHMs. The expansion of red and orange areas, particularly along the southern river system and the western valleys, indicates that flood risk will increase even in this conservative climate prediction. This is a critical point for planning demining operations, as future flood risks must be accounted for to ensure safety in affected areas.

Under the SS245 moderate-emissions scenario, shown in Figure 33, the flood risk areas expand further compared to the SS126 trajectory. The moderate to high-risk zones (yellow and orange) grow and relocate depending on the month. Some of the very high-risk zones reduce in size, however, generally this reduction is replaced by an expansion of moderate and high-risk zones. This is indicative of worsening flood conditions due to higher predicted precipitation levels under this intermediate emissions scenario.

Figure 34 presents the FHM under the SS585 high-emissions scenario, representing the most severe climate change trajectory. Generally, the flood hazard risks are considerably higher in this model, with red and orange zones covering a larger portion of the study area compared to the SS126 and SS245 scenarios. Compared to the SS126 FHM the southern river system exhibits a significant expansion of flood-prone areas during the month of November. Interestingly, however, the FHM for May shows some improvement compared to the SS245 scenario, suggesting that precipitation patters may be more favorable during this month under high-emissions trajectories. This anomaly may be due to variability in seasonal precipitation distribution or inaccuracies of the model's predictions, requiring further analysis for a definitive explanation.

When comparing the three scenarios, the general trend shows a worsening of flood conditions across almost all months, though the differences between SS245 and SS585 are not as drastic as might be expected. The largest increases in flood-prone areas are observed

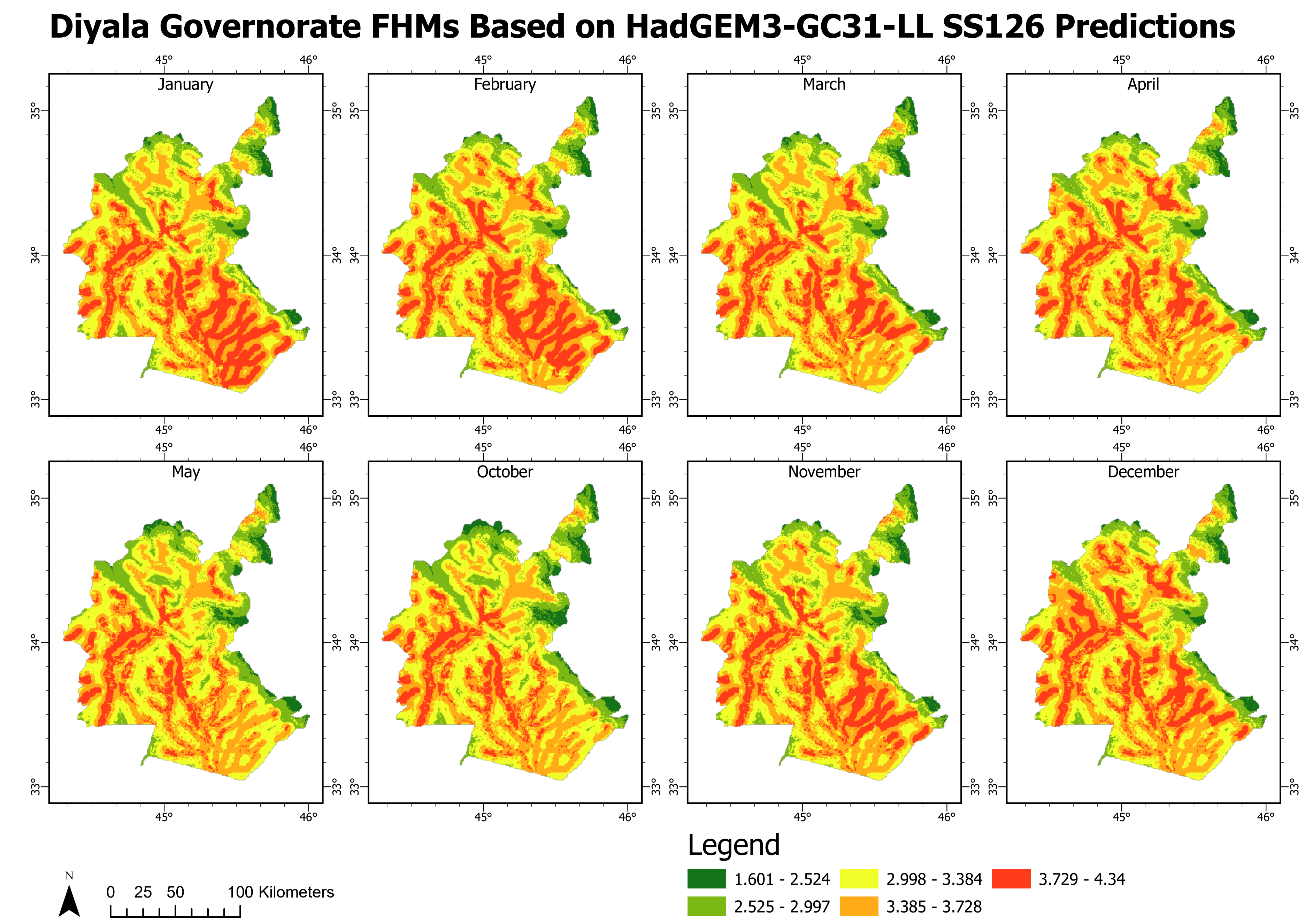

Figure 32: Flood Hazard Maps (FHMs) based on SS126 trajectory precipitation predictions from the CMIP6 model. The SS126 scenario represents a low-emissions trajectory, projecting minimal increases in precipitation levels. The map illustrates the anticipated flood hazard risks for the study area from October to May, with risk categories ranging from very low (green) to very high (red). While the SS126 scenario is the most optimistic of the models, a noticeable increase in flood-prone areas is observed when compared to historical data.

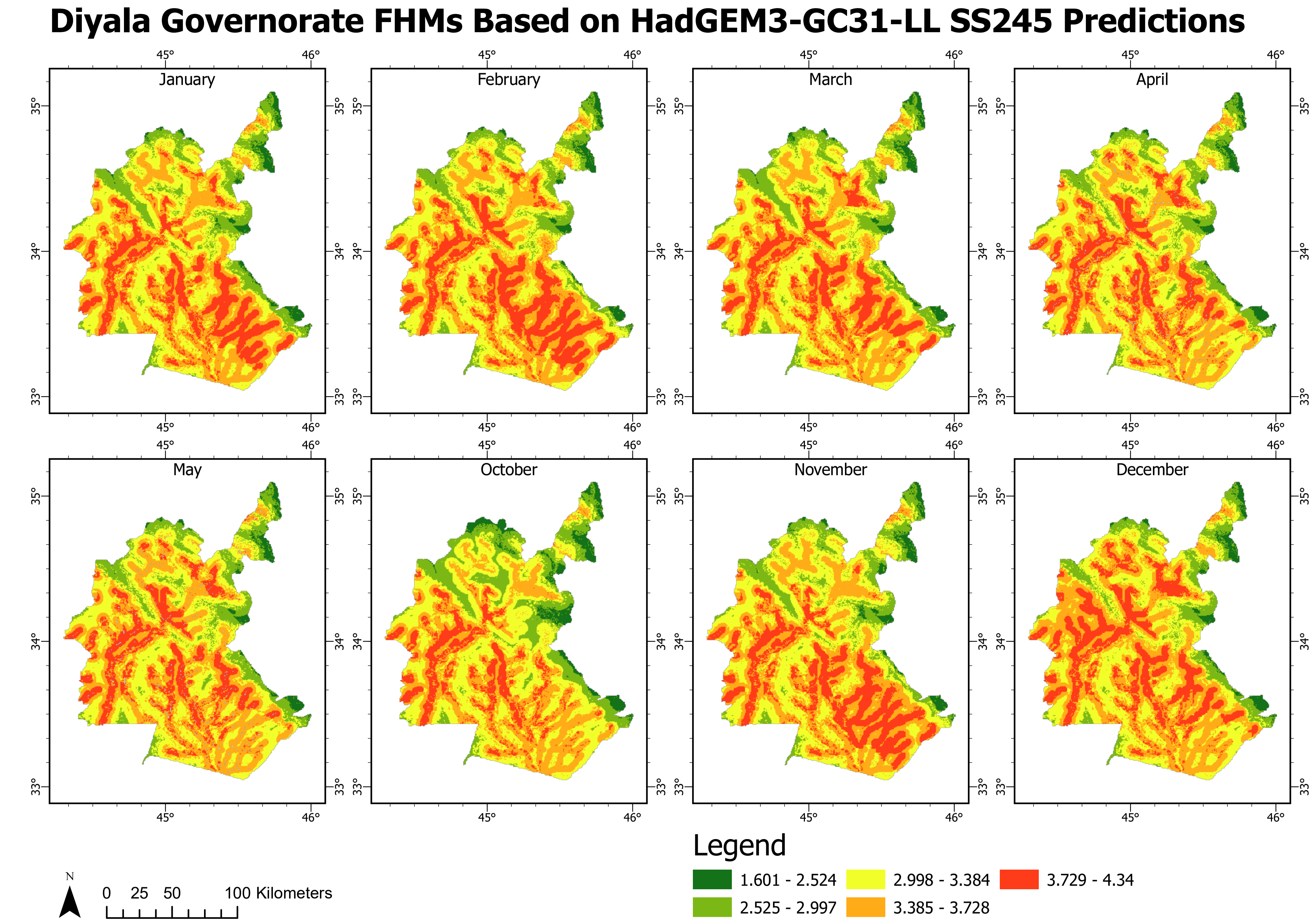

Figure 33: Flood Hazard Maps (FHMs) based on SS245 trajectory precipitation predictions from the CMIP6 model. The SS245 scenario reflects a moderate-emissions trajectory with a corresponding rise in projected precipitation levels. The map shows the predicted flood hazards for the region during the critical months of October to May, classified into five risk levels. A considerable expansion of high-risk zones (orange and red) can be seen, particularly in the southern river system and along the western valleys, indicating heightened flood risks compared to the SS126 scenario.

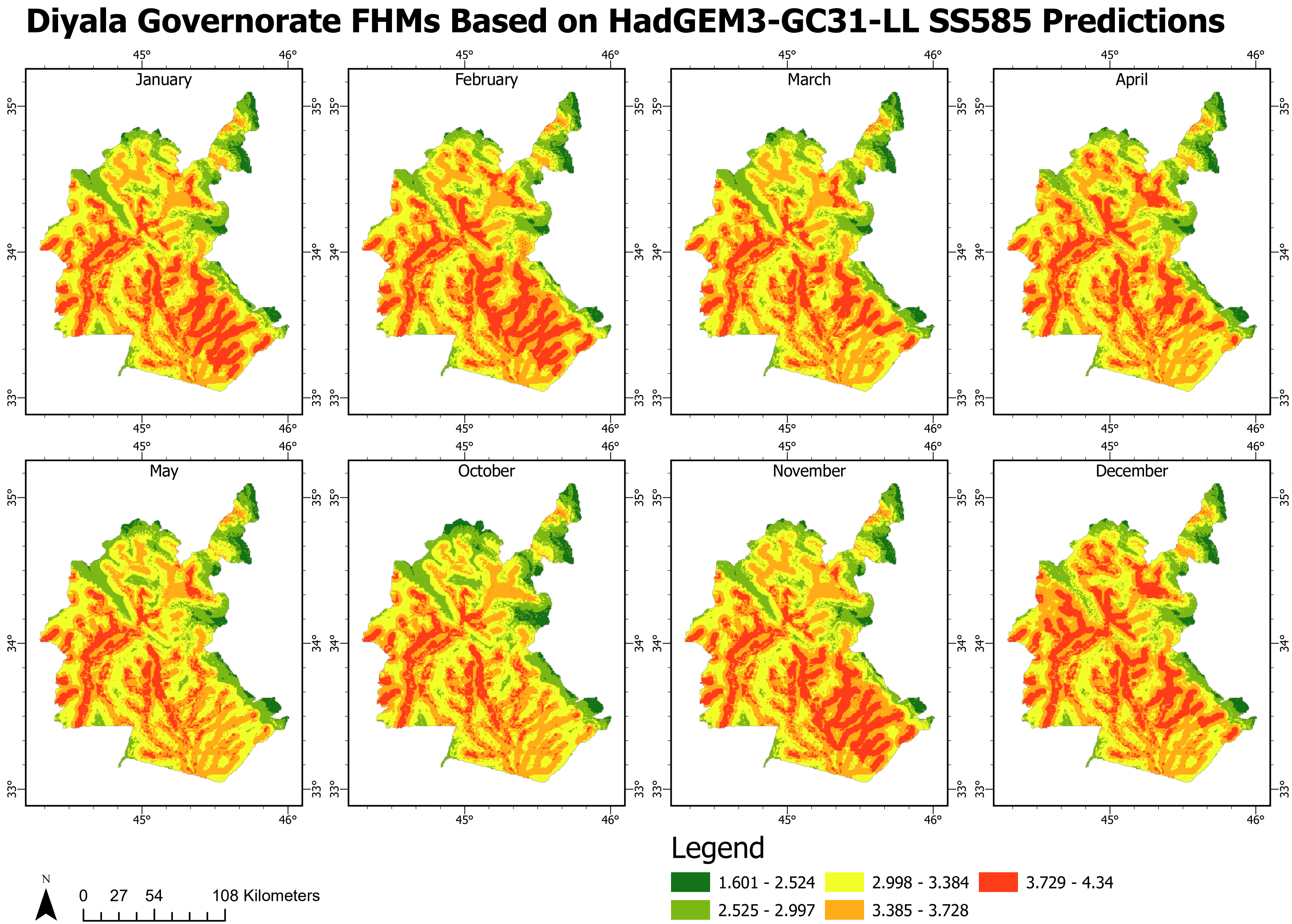

Figure 34: Flood Hazard Maps (FHMs) based on SS585 trajectory precipitation predictions from the CMIP6 model. The SS585 scenario represents a high-emissions trajectory, projecting the most significant increases in precipitation levels. This map presents the anticipated flood hazards for the study area from October to May, with risk categories ranging from very low (green) to very high (red). The SS585 scenario shows a dramatic increase in flood-prone areas, especially in the southern and western regions, marking a significant intensification of flood risk compared to both SS126 and SS245 scenarios.

Table 15: Minefields in Diyala Affected By Flooding

Mine ID	Latitude	Longitude	Flood Risk
ID-1708	34.254411	45.209587	low
ID-1709	34.254411	45.209587	low
ID-1710	34.254411	45.209587	low
ID-1711	34.254411	45.209587	moderate
ID-1712	34.254411	45.209587	low
ID-1713	34.254411	45.209587	moderate
ID-1714	34.254411	45.209587	low
ID-2529	34.311363	45.103783	moderate
ID-2530	34.311363	45.103783	low
ID-2531	34.311363	45.103783	moderate
ID-2532	34.311363	45.103783	moderate
ID-2533	34.311363	45.103783	moderate
ID-2534	34.311363	45.103783	moderate
ID-2535	34.311363	45.103783	moderate
ID-2536	34.311363	45.103783	moderate
ID-2537	34.311363	45.103783	low
ID-2549	34.250662	45.160107	very high

between SS126 and SS245, indicating that even moderate levels of emissions could lead to noticeable changes in flood risks. Future work could also consider the SS370 scenario, which was unavailable for the HadGEM3-GC31-LL model at the time of this analysis.

In summary, the predictive FHMs highlight the increasing risk of flooding under all future climate scenarios. The SS126 model, while representing the most optimistic outcome, still shows a clear increase in flood hazards compared to historical conditions. The more severe SS245 and SS585 scenarios indicate that future flooding will likely become a major concern, particularly in critical areas for demining efforts. As flooding worsens, the planning and execution of these operations will need to adapt to the increased risks, ensuring both the safety of personnel and the effectiveness of demining activities in the region.

5.4 Comparison of Hazardous Areas

In this section, the distribution of known hazardous areas, specifically uncleared minefields from the Hazard Area Dataset provided by UNMAS, are compared against the flood risk zones identified by the FHMs. The specific mines affected by flooding and their corresponding risk levels are detailed in Table 15. As shown in Figure 35, several minefields are located within high-risk flood zones. This alignment of hazardous areas with flood-prone regions emphasizes the importance of incorporating flood risk data into the planning of demining operations.

When examining the flood hazard classifications across the different FHM sets in Figure 35, the initial FHM (Section 5.1) classifies all minefields into a higher flood risk category compared to the FHMs from subsequent sets. Notably, one minefield in particular, designated as Mine ID-2549, shifts from a very high-risk (red) zone in the first FHM set to a high-risk (orange) zone in the later FHMs. Despite this reclassification, it remains a zone of significant flood risk. Therefore, this mine should be prioritized for clearance due to its persistently high risk of flooding.

The monthly analysis, as visualized in Figure 36, shows that the flood risk levels associated with these hazardous areas remain relatively static over time, with minimal seasonal fluctuations. For instance, Minefield ID-192 (including the mines ID-2529, ID-2530, ID-2531, ID-2532, ID-2533, ID-2534, ID-2535, ID-2536, ID-2537) experiences a reduction in flood risk, moving from a moderate-risk (yellow) to a low-risk (green) zone during the month of May. Despite this temporary decrease in risk, the overall flood hazards associated with the minefields remain significant, and no major increase in flood risk is observed across the other months, allowing for more predictable planning of demining operations.

The zoomed-in analysis in Figure 37 provides further clarity on the flood risks facing the individual minefields. The Mine ID-2549, located in the orange high-risk zone, should be the first priority for demining efforts, given its persistent vulnerability to flooding. This should be followed by the minefield ID-192, which is situated in a moderate-risk (yellow) zone. Lastly, the minefield ID-130 located within a mixed yellow green zone, should be addressed as its flood risk fluctuates but remains relatively manageable.

Overall, the alignment of flood hazard data with known hazardous areas enhances the strategic planning of demining operations by identifying which minefields are most at risk of flooding and should be prioritized for clearance. As the flood risk for these areas remains largely consistent, operations can be planned and prioritized with greater confidence in the stability of flood risk levels.

Several variables may have influenced the results, including the specific criteria selected for the AHP analysis and the relative weightings assigned to each criterion. Additionally, the spatial resolution of the input data and the hydrological models employed could impact the accuracy and granularity of the FHMs.

From Table 15, it becomes apparent that mines belonging to one larger hazardous area are grouped under a single coordinate location. To further enhance demining planning, it would be beneficial to determine the exact coordinates of individual mines within these broader hazardous areas. This refinement would improve the precision of flood risk assessments and the prioritization of demining efforts.

One limitation of this case study is the inability of the FHMs to accurately capture the flood hazard posed by Hamrin Lake shown in blue in Figure 29. This discrepancy arises from the lack of up-to-date data on the current extent of the lake at the time of analysis, likely due to seasonal variations or incomplete hydrological records. To address this gap, future analyses should incorporate more localized and current datasets, such as detailed water body maps or shapefiles accurately representing the lake's extent, to improve the precision of flood hazard assessments in its vicinity.

However, despite this limitation, the overall flood risk analysis remains valid. The primary hazardous areas identified are located outside the lake's immediate floodplain, and the lake's flood dynamics are largely governed by the confluence of rivers feeding into it, which are already captured in the model. Thus, while future enhancements could refine the assessment, the current analysis provides a robust foundation for identifying flood-prone areas.

5.5 Automated Generation of Flood Hazard Maps

To streamline the process of FHM generation, a set of custom tools was developed as part of this case study. This system, depicted in Figure 38, offers a more efficient and user-friendly alternative to the initial workflow outlined in the methodology. The approach is inspired by the work of Ekeanyanwu et al., who created an ArcGIS tool for flood hazard modeling and mapping in Austin, Texas. The goal was to create a similar tool, with enhancements such as integrating the calculation of weights using the AHP method, to make

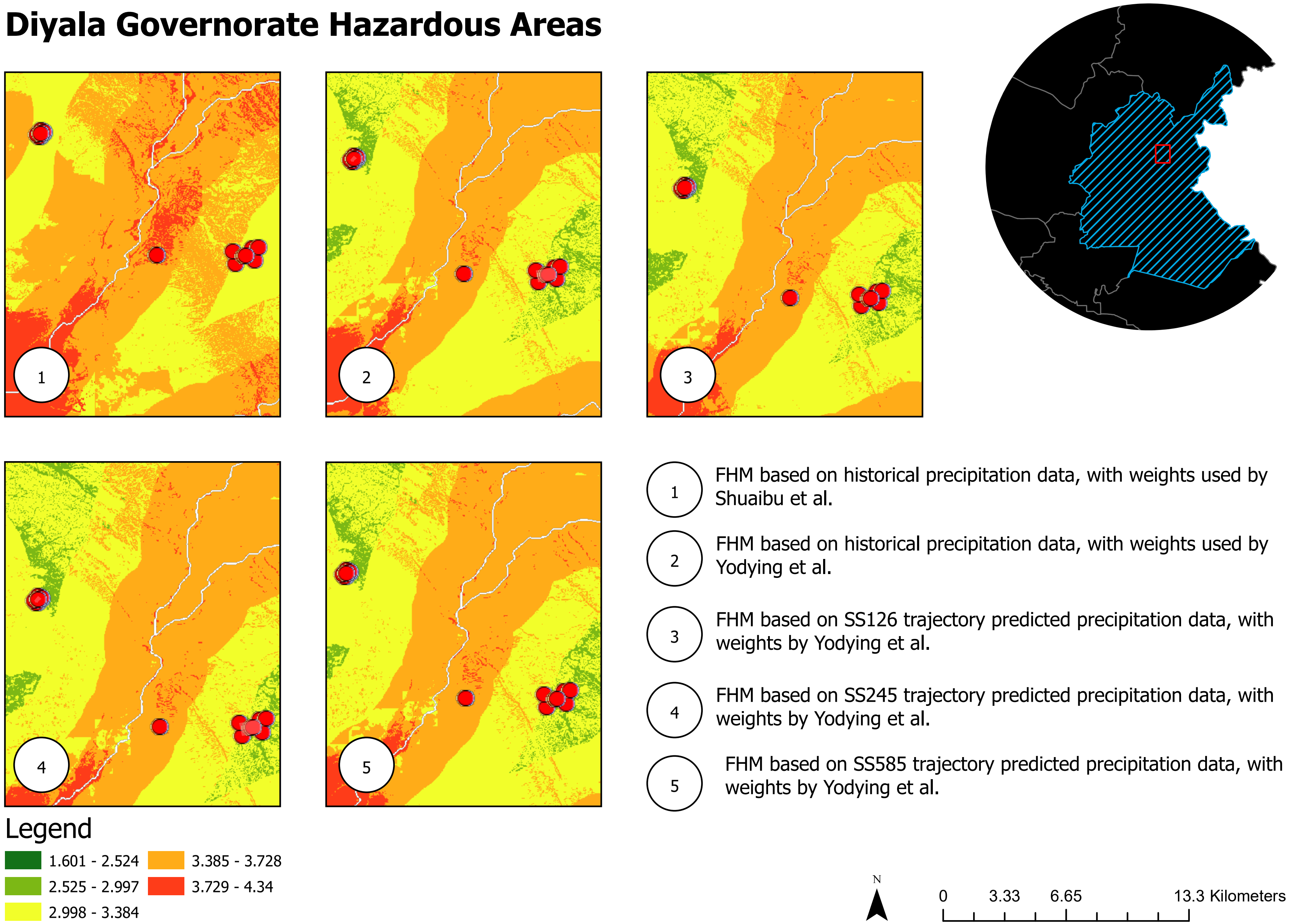

Figure 35: Overview of Hazardous Areas across All Flood Hazard Maps (FHMs). This figure displays the distribution of hazardous areas in relation to flood-prone zones across all FHMs, highlighting areas at varying levels of flood risk.

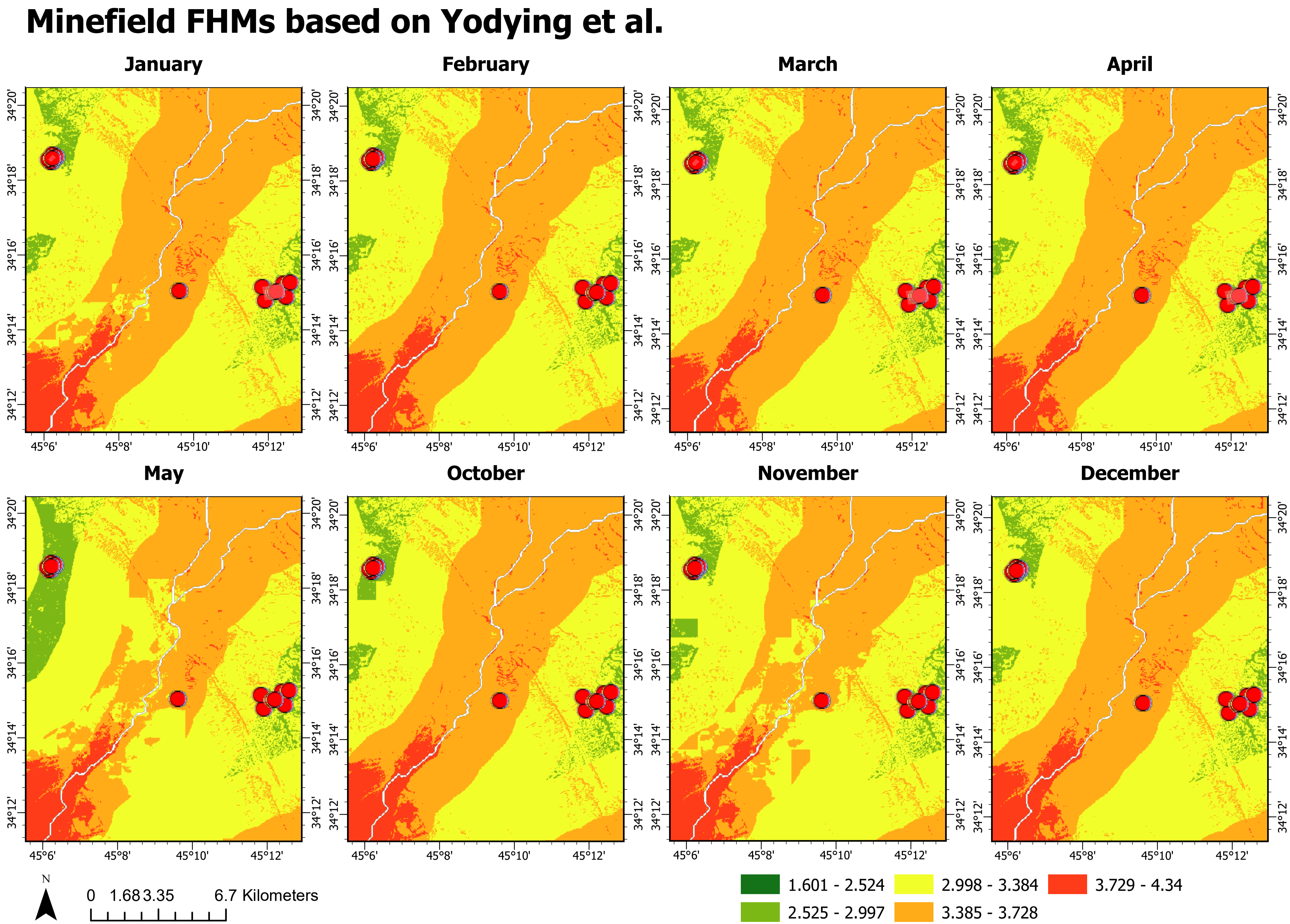

Figure 36: Zoomed Overview of Hazardous Areas Across All Months. This figure provides a closer look at the relationship between known hazardous areas and monthly flood hazard levels, identifying fluctuations in flood risk across different months.

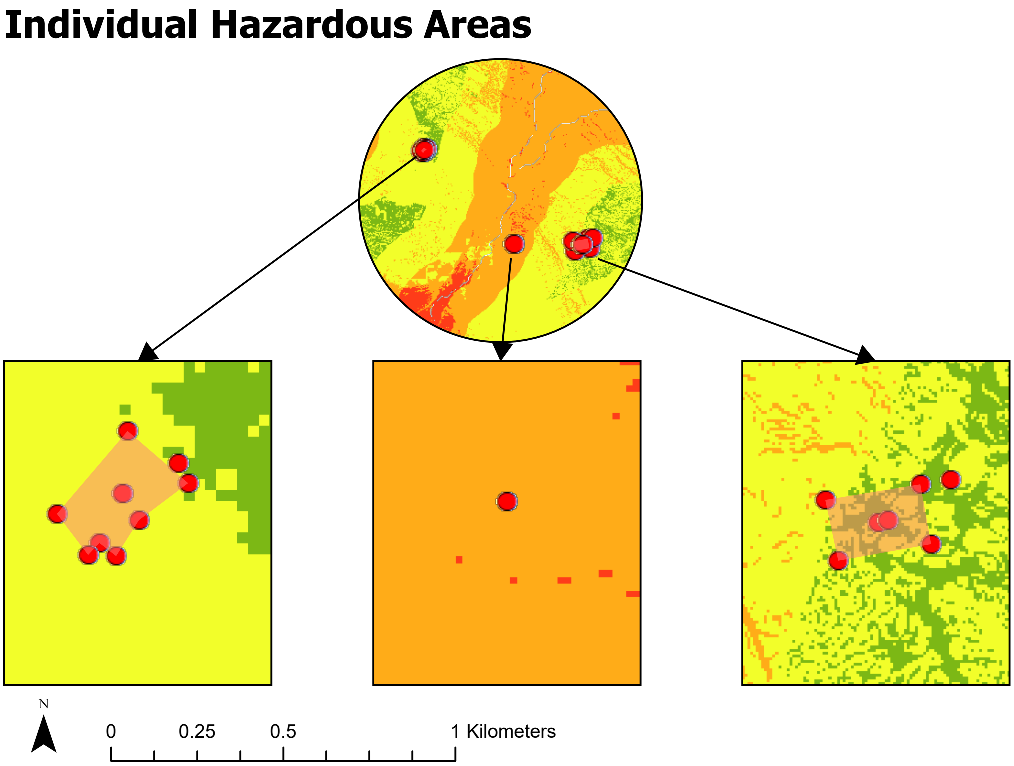

Figure 37: Detailed View of Individual Hazardous Areas. This zoomed-in figure highlights specific hazardous areas in relation to flood risk levels, focusing on particular minefields and their associated flood hazards. From left to right: the fir st image displays Minefield ID-192, the second image shows Mine ID-2549, and image three presents Minefield ID-130

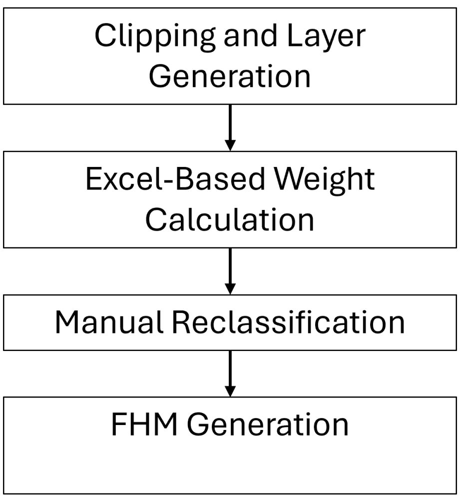

Figure 38: Automated FHM Generation Process Flow Chart

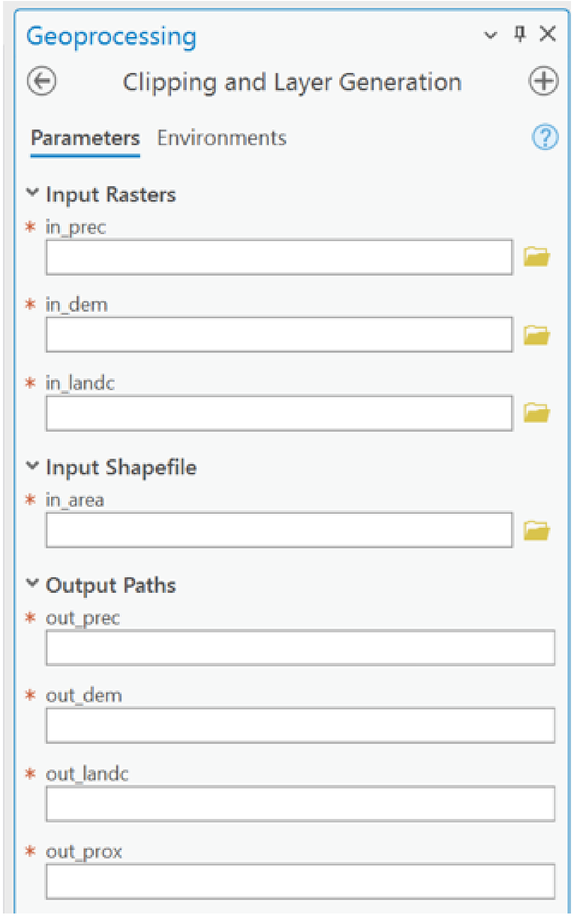

Figure 39: Custom Clipping and Layer Generation Tool

the entire workflow more accessible to GIS users and practitioners. This system consists of two custom-built tools within ArcGIS Pro, accompanied by an Excel-based weight determination table. Together, these components significantly reduce manual intervention and provide flexibility to modify the tools for specific project needs.

5.5.1 Preprocessing and Layer Generation Tool

The first tool automates the data preparation and DEM modification process, where several key geospatial layers are derived from four inputs: the DEM, land cover, and precipitation rasters, along with a shapefile representing the study area. As shown in Figure 39, the user interface of Tool 1 is designed to be intuitive, with asterisks marking required inputs and blue info markers providing further guidance on each data field.

This tool processes the inputs by cropping them to the study area's boundaries and generating critical layers, including slope, fill, flow direction, flow accumulation, streams, and stream proximity. The outputs are saved at a user-specified location, forming the foundation for subsequent flood hazard analysis. The automation of this step significantly reduces repetitive manual work and accelerates the preparation of datasets for further analysis.

5.5.2 Flood Hazard Map Generation Tool

The second tool is responsible for performing the Weighted Sum Analysis to generate the final FHM. It takes five pre-reclassified raster layers — DEM, land cover, precipitation,

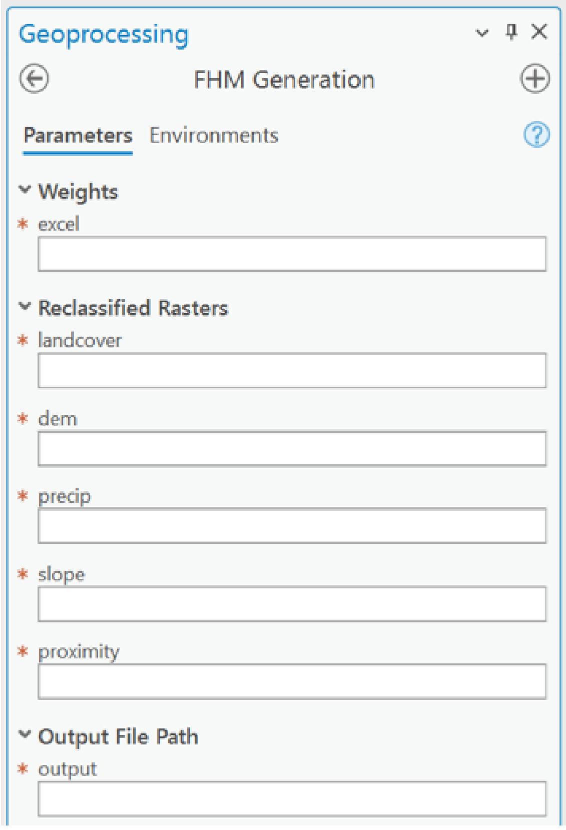

Figure 40: Custom FHM Generation Tool

slope, and stream proximity — and combines them using weights calculated through the AHP method. The relative importance of each input in influencing flood risk is determined by these weights, which are computed and stored in an Excel worksheet. The user interface of Tool 2 is shown in Figure 40, allowing users to upload their reclassified rasters and specify the Excel file containing the weights.

The Excel-based tool plays a critical role in the workflow. Users provide a comparison matrix through which the AHP method calculates the weights for each input layer. The tool also checks the CR to ensure it meets the acceptable threshold of 0.1. If the CR is valid, the relevant field turns green, indicating that the weights are appropriate for use in the analysis. The required user inputs in the Excel tool are marked in blue for easy identification.

Once the weights are calculated and imported into Tool 2, they are applied during the weighted sum analysis, producing a comprehensive FHM. Each input layer's contribution to the final map is weighted according to user-defined preferences, ensuring that the tool remains accessible even to non-experts. This step automates much of the map creation process, making it suitable for a broad range of users.

5.5.3 Limitations and Potential Improvements

While this automated toolset substantially reduces the effort required for FHM generation, some limitations remain. The most significant challenge is that users must manually reclassify the input layers (DEM, land cover, precipitation, slope, and stream proximity) before running the weighted sum analysis. This reclassification is crucial for standardizing the datasets on comparable scales. However, the use of the Natural Breaks (Jenks) method, which is optimal for dividing data into meaningful intervals, is not supported by the

ArcPy library for raster datasets. Therefore, users must perform this step manually within ArcGIS Pro, as described in Section 4.2.

Additionally, the manual classification of the land cover layer remains a bottleneck in the automation process. Future updates to the toolset could address these limitations by incorporating advanced classification methods directly within the automated workflow. This would allow for a fully automated process, from initial input to the final FHM output, minimizing manual intervention and further improving user experience.

6 Future Work

The application of satellite imagery, particularly *Synthetic Aperture Radar (SAR)*, holds significant potential for enhancing flood risk assessments (Esri, 2024i). Since the launch of SEASAT in June 1978, SAR has been instrumental in providing detailed information on various oceanic phenomena, including surface waves, currents, and upwellings (Jackson et al., 2004). SAR's ability to detect even minute changes in surface roughness, combined with its capability to operate under any weather conditions and at any time of day, makes it exceptionally well-suited for identifying water bodies and flooded areas, especially in regions where cloud cover might obstruct optical imagery (Jackson et al., 2004).

SAR works by emitting pulses of microwave energy, which travel to Earth's surface and are then reflected back to the satellite's antenna. The time delay between the outgoing and incoming signals enables precise distance measurements, with range resolution improving as pulse bandwidth increases. Typical SAR systems achieve resolutions between 15 and 3.7 meters for bandwidths of 10 to 40 MHz (Jackson et al., 2004). Additionally, SAR allows for control over signal polarization, enhancing its ability to distinguish between surface features. For flood risk mapping, co-polarized data, particularly horizontal polarization, is effective in creating high contrast between water surfaces and surrounding terrain (Amitrano et al., 2024). This data can also be integrated with fuzzy classifiers and machine learning techniques to accurately delineate flooded regions, as demonstrated by Amitrano et al.

Beyond SAR applications, refining the weighting of flood risk factors remains a critical area for future research. Collaboration with local experts, who possess in-depth knowledge of the region's hydrology, topography, and historical flood patterns, is essential for ensuring the accuracy of the weighting process. Historical flood data can also be leveraged to refine these weights, improving the model's ability to predict flood hazards with greater precision.

Exploring alternative weighting methods, such as Frequency Ratio or Shannon's Entropy, could also enhance the accuracy of FHMs. These approaches provide a more systematic framework for integrating multiple criteria, reducing uncertainties and minimizing subjective bias in the analysis (Roopnarine et al., 2022). In addition, machine learning algorithms could be employed to optimize the weights automatically based on historical flood data, leading to more adaptive and precise flood risk predictions.

Expanding the range of flood hazard criteria is another key step to improving analysis. Factors such as soil type, drainage density, and proximity to infrastructure should be considered in future studies (Shuaibu et al., 2022). An important area for exploration is the role of mine weight and size in flood scenarios. Assessing how these factors influence the distance mines might travel during inundation could enhance predictions of mine mobility, leading to more accurate hazard assessments. In summary, future advancements in flood hazard mapping should focus on leveraging satellite technologies like SAR, refining weight assignment methodologies, and expanding the range of hazard criteria, including the consideration of mine mobility during floods.

7 Conclusion

This thesis demonstrates the critical importance of integrating flood hazard mapping with demining operations in regions like the Diyala Governorate, Iraq, where the risks posed by both landmines and seasonal flooding intersect. The case study's application of GIS and AHP enables the creation of flood hazard maps that offer a detailed spatial analysis of areas most vulnerable to flooding and mine displacement. These maps allow for a more strategic approach to demining, prioritizing high-risk zones and improving the allocation of resources. While the case study highlights the efficacy of this approach, it also points to areas for future research, including the refinement of flood hazard criteria and the incorporation of more localized, up-to-date data on hydrological and minefield conditions. Ultimately, this research lays the groundwork for more efficient and safer demining operations, with the potential to mitigate the dangers faced by civilians and demining personnel in post-conflict settings. By bridging the gap between flood risk analysis and mine clearance, this work contributes to the broader efforts of post-conflict recovery and sustainable development in Iraq.

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Acronyms

AHP Analytical Hierarchy Process 5, 15, 28, 41, 46

ALOS Advanced Land Observing Satellite 11

APII Anti-Personnel Mine Ban Convention 3

CI Consistency Index 29, 30, 32

CMIP6 Coupled Model Intercomparison Project 6 9

CR Consistency Ratio 29–32, 46

D8 Deterministic 8-neighbor 19, 21

DEM Digital Elevation Model 4, 5, 11, 15, 17–19, 21, 34, 45

DINF D-Infinity 21

FHM Flood Hazard Map 4, 6, 9, 10, 14, 29, 31, 32, 34, 36, 40, 41, 45–47

GCM General Circulation Model 9, 10

GIS Geographic Information System 4, 5, 12

ICBL International Campaign to Ban Landmines 3

KML Keyhole Markup Language 12

MCDM Multi-Criteria Decision-Making 5

MFD Multiple Flow Direction 21

NGO Non-Governmental Organization 3

PRISM Panchromatic Remote-sensing Instrument for Stereo Mapping 11

RCP Representative Concentration Pathway 10

RI Random Index 29, 30

SAR Synthetic Aperture Radar 47

SSP Shared Socioeconomic Pathway 9, 10

UNDP United Nations Development Programme 10

UNMAS United Nations Mine Action Service 3, 12

UXO Unexploded Ordnance 3